

Demographic Transition and Structural Transformation*

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September 14, 2026

Abstract

We explore the effect of demographic transition on structural transformation. When fertility declines, a larger share of the population may remain in farming due to agriculture's reliance on a fixed factor of production, land. We test this hypothesis leveraging a quasi-experimental family planning and child health program provided to Bangladeshi households. Driven by land congestion, the program raises the agricultural employment share decades later. Improving human capital can offset the effect of fertility declines on the agricultural employment share. Despite slower structural transformation, the program raises per-capita expenditure.

Keywords: Fertility, human capital, industrialization, economic growth.

JEL Categories: O1, O41, J1.

*We are grateful to David Canning, Yuzuru Kumon, Mushfiq Mobarak, Suresh Naidu, Francisco Javier Rodriguez Román, and Frédéric Robert-Nicoud for helpful discussions, and appreciate comments from Wookun Kim, Doug Gollin, Adriana Lleras-Muney, David Weil, and Asger Wingender. We thank seminar participants at UCLA, Collegio Carlo Alberto, the University of San Francisco, the European University Institute, and the Free University of Bozen-Bolzano, as well as participants at numerous conferences and workshops for helpful comments. We are grateful to Antonio Curcio, Ilaria Malisan, Jessica Mancuso, Alec Truax, Sara Truesdale, and Elizabeth Sorensen Montoya for research assistance. We acknowledge financial support from CEPR STEG. McCully acknowledges financial support from CCPR's Population Research Infrastructure Grant P2C from NICHD: P2C-HD041022, CCPR's Population Research Training Grants T32 from NICHD: T32-HD007545, and the Institute on Global Conflict and Cooperation. The data collection in Bangladesh was generously funded by the National Institutes of Health, the Population Research Bureau, and the International Initiative for Impact Evaluation. All errors are our own.

1 Introduction

Economic growth is characterized by two fundamental processes: the demographic transition, marked by declining fertility and mortality, and structural transformation, in which workers shift from agriculture to manufacturing and services. While a large literature explores how economic development and structural transformation influence fertility and mortality patterns (Galor and Weil 1996, 2000; Chatterjee and Vogl 2018; Ager et al. 2020), less is known about how demographic change affects structural transformation. As fertility declines in virtually every country (Delventhal et al. 2021), and with global population expected to decline within the next 60 years (United Nations 2024), understanding the role of falling fertility on structural transformation is increasingly vital.

Theoretically, the effect of demographic transition on structural transformation is ambiguous. In Malthusian models where agriculture relies on land—a fixed factor of production—a smaller labor force reduces land congestion, increases returns to labor in farming, and slows the reallocation of labor out of agriculture (Malthus 1798; Lewis 1954). Land congestion may be particularly relevant in open economies, where food imports can satisfy domestic demand and dampen demand-side pressures. At the same time, the demographic transition may promote human capital accumulation. Lower fertility facilitates quantity–quality substitution (Barro and Becker 1989), while improved child survival increases the expected returns to investment in each child (Cervellati and Sunde 2005). If nonagricultural sectors benefit disproportionately from skilled labor, rising human capital may accelerate structural transformation. Whether Malthusian land congestion or human capital accumulation dominates is therefore an empirical question.

We formalize this intuition within a stylized two-sector overlapping-generations model, in which reallocation out of agriculture raises wages due to a wage wedge that preserves inefficiently high agricultural employment, in line with Gollin et al. (2014). Empirically testing the model and evaluating which mechanism dominates presents several challenges. First, fertility may be endogenous to economic development and the sectoral composition of the labor market. Second, the effects of fertility decline on labor supply emerge only gradually, as affected cohorts age into the workforce, requiring data across decades. Third, identifying underlying mechanisms, particularly the role of human capital, necessitates individual- or household-level data to credibly link intermediate outcomes to sectoral shifts.

We address these challenges by leveraging a landmark family planning and child health program in rural Bangladesh that reduced fertility and child mortality, along with rich panel data collected across four decades with exceptionally low attrition. Launched in 1977, the program introduced modern contraception and later expanded to preventive child health

services, such as immunizations, that improved adolescent human capital ([Barham 2012](#)). By reducing fertility and improving child survival, the program accelerated the demographic transition ([Phillips et al. 1982](#); [Joshi and Schultz 2007](#)). Evaluation was built into the design of the program with treatment assigned at the village level and strong baseline balance between treated and comparison areas.

We exploit the quasi-randomized placement of the Maternal and Child Health and Family Planning (MCH-FP) program to estimate intent-to-treat (ITT) effects of demographic transition on structural transformation, and its underlying mechanisms—land congestion and human capital. Thirty-five years after program onset, household-level data show that treated households allocate a 19 percent higher share of labor to agriculture and a 12 percent lower share to manufacturing, indicating that the land-congestion mechanism dominates. Consistent with this interpretation, IV-scaled estimates imply that program-induced reductions in household young adult male cohort size are associated with substantially higher agricultural work shares, while total agricultural labor in levels changes little—suggesting labor input is effectively fixed relative to available land. Eligible households also send fewer migrants to urban areas, providing complementary evidence on how the slower reallocation out of agriculture unfolded.

To examine the human-capital channel, we exploit cohort-level variation in exposure to the MCH-FP’s intensive child health phase using individual-level data. Men born in treatment areas during the intensive child health phase perform better on cognitive tests in childhood and attain more schooling than similarly aged men in comparison areas ([Barham 2012](#); [Barham et al. 2025](#)). The findings in this paper further demonstrate that this cohort is more likely to work outside agriculture—particularly in the service sector, where returns to skill are higher—indicating that human capital can partially offset Malthusian land congestion. These sectoral shifts are accompanied by modest adjustments in farming practices, including movement toward less labor-intensive crops, consistent with a rising opportunity cost of farm labor.

Whether this slower structural transformation reduced material living standards is theoretically ambiguous. Fewer children and greater human capital tend to raise expenditure per person, while greater employment in lower-paid agriculture tends to reduce it. We estimate the program raised expenditure per capita by 11%.

This paper makes three contributions to the literature on the consequences of fertility decline for economic growth ([Ashraf et al. 2013](#); [Cavalcanti et al. 2021](#); [Jones 2022](#); [Hopenhayn et al. 2022](#)). First, we provide novel causal evidence on the effect of demographic transition on structural transformation, two central processes of economic development ([Kuznets 1957](#)).¹

¹[Li and Zhang \(2007\)](#) estimate fertility’s effect on growth using minority share as an instrument; their

While a large literature studies how structural transformation and productivity growth affect fertility and the demographic transition (Greenwood and Seshadri 2002; Wanamaker 2012; Ager et al. 2020), fewer studies investigate the reverse relationship: how population growth and fertility decline shape structural transformation. Voigtländer and Voth (2013) and Leukhina and Turnovsky (2016) study the role of population size in structural transformation,² while Cheung (2023) emphasizes the quantity-quality tradeoff; each of these three studies focus on historical contexts with limited data. Gollin and Rogerson (2014) and Herrendorf et al. (2012) quantitatively explore the role of transportation infrastructure facilitating population movements and thereby structural transformation. However, the above papers rely on calibrated macroeconomic models, inhibiting causal identification of the effects of fertility on structural transformation and of the corresponding mechanisms at the household level.³

Second, we develop a theoretical model which explicitly considers endogenous fertility featuring the quantity-quality tradeoff and family planning with multiple production sectors, allowing an investigation of the structural transformation effect of family planning interventions. The few growth models jointly featuring endogenous fertility, the quantity-quality tradeoff, and family planning technologies do not include multiple sectors, nor a fixed factor of production (Strulik 2017; Cavalcanti et al. 2021). As a result, the Malthusian land congestion mechanism and structural transformation that we emphasize is absent in those models.

Third, we show that land, a fixed factor of production, can fundamentally shape how fertility decline affects structural transformation. While unified growth theory incorporates land congestion with endogenous fertility, technological progress ultimately overcomes this Malthusian force (Galor and Weil 2000; Galor 2005). We show that land congestion remains a quantitatively important determinant of the sectoral response to fertility decline in a modern open economy.

2 Model

We present a parsimonious model of structural transformation with two sectors—agriculture and manufacturing—and two factors of production, land and labor. Overlapping genera-

result that lower birth rates hamper growth requires minority share to affect growth only through fertility.

²Voigtländer and Voth (2013) abstracts from human capital, and therefore the quantity-quality tradeoff in their model, which they apply to the pre-modern period. Leukhina and Turnovsky (2016) assume exogenous population. Relative to both papers, our model allows for endogenous population growth, a quantity-quality tradeoff, and a role for family planning technologies to affect fertility.

³Fertility and agricultural employment may comove due to confounding factors such as changes in skill-biased technical change, which alter the returns to child quality investments and to employment in agriculture.

tions live together in households where adults decide the quantity and education of children. Households derive utility from sexual activity and can reduce the probability of births by purchasing contraception. We assess how lower contraception costs and higher child survival affect fertility, human capital, the agricultural employment share, and per capita consumption.

2.1 Setup

2.1.1 Production

Consider a small open economy that trades agricultural and manufacturing goods with the world economy.⁴ In total there are T units of land, which are only used in agriculture.

Production of agricultural output is Cobb-Douglas:

$$Q_{at} = A_{at} L_{at}^{\theta} T_{at}^{1-\theta}, \quad (1)$$

where Q_{at} is the quantity of agricultural output in period t , A_{at} is Hicks-neutral agricultural productivity, L_{at} is the quantity of labor employed in agriculture, and T_{at} is the quantity of land used in agriculture (equal to T in equilibrium). $\theta \in (0, 1)$ is the labor income share in agriculture. Land rents are paid to absentee landlords.

Production in manufacturing is linear in labor:

$$Q_{mt} = A_{mt} h_t L_{mt}, \quad (2)$$

where Q_{mt} is the quantity of manufacturing output, A_{mt} is Hicks-neutral manufacturing productivity, and L_{mt} is the quantity of labor employed in manufacturing.⁵ As in [Caselli and Coleman \(2001\)](#) and [Porzio et al. \(2022\)](#), per household human capital h_t only yields returns outside of agriculture.⁶

In many developing economies, employment is inefficiently high in agriculture ([Gollin et al. 2014](#); [Munshi and Rosenzweig 2016](#)). To capture this feature in our model, we assume

⁴The small open economy assumption implies prices are exogenous and unaffected by local demand. See Section 2.3 and A.4 for discussion of adding trade costs. Table 2 shows that the program we study induced only modest changes in consumption shares across sectors, suggesting demand-side factors do not drive sectoral labor reallocation in our empirical setting. Relatedly, [Cavalcanti et al. \(2021\)](#) find in a model calibrated to Kenya that the effects of family planning interventions are similar when general-equilibrium price adjustments are shut down.

⁵The linear manufacturing production function assumption is isomorphic to a small open economy with freely mobile capital; see Online Appendix A.1.

⁶A less restrictive assumption would allow human capital to boost output in both sectors, but more so in manufacturing. Doing so does not change the main predictions of the model.

that labor markets are distorted by a wage wedge, such that agricultural wages are lower than nonagricultural wages:

$$w_{at} = \xi w_{mt}, \quad (3)$$

where $\xi \in (0, 1)$.

2.1.2 Households

Households are made up of adults and children.⁷ To characterize household preferences, we build on [Strulik \(2017\)](#). Utility is defined as

$$U = \log c_{at} + \delta \log c_{mt} + \alpha \log(\pi_t n_t) + \gamma \log h_{t+1} + \sigma \log s_t,$$

where c_{at} and c_{mt} are consumption of agricultural and manufacturing goods. $\pi_t \in (0, 1]$ is the child survival rate, n_t is births per household, so $\pi_t n_t$ is net fertility. h_{t+1} is children's next-period human capital, and s_t is sex.

Births $n_t = \min\{s_t - \mu u_t, \bar{n}\}$ are the minimum of sex s_t net of contraception-prevented births and the biological maximum reproduction $\bar{n}$; in what follows, we consider only interior solutions. μ is the effectiveness of the family planning technology such that a unit of contraception u_t prevents the birth of μ children. Sex is proportional to births according to some constant that we normalize to 1.

Human capital is produced according to

$$h_{t+1} = A_{ht} h_t e_{t+1}^\eta,$$

where A_{ht} is exogenous human capital production productivity, e_{t+1} is the time spent by adults educating each child, and $\eta > 0$ governs the returns to education. We assume $\alpha > \eta\gamma$ to ensure an interior solution with positive fertility even when births can be costlessly avoided. Households have one unit of time and therefore face the budget constraint

$$\bar{w}_t [1 - (\phi + e_{t+1})\pi_t n_t] = p_{ft} u_t + p_{at} c_{at} + p_{mt} c_{mt},$$

where ϕ denotes per surviving child rearing time and p_{ft} is the price of a unit of the family planning technology. $\bar{w}_t \equiv \frac{w_{mt} L_{mt} + w_{at} L_{at}}{L_t}$ denotes the mean wage. The price of agricultural goods is p_{at} and of manufacturing goods p_{mt} . Each household works a fraction of their time endowment equal to $\ell_t = 1 - (\phi + e_{t+1})\pi_t n_t$. Aggregate labor supply L_t is a product of net

⁷Each household may contain one or more adults. Our model assumes a unitary household, in which everyone aims to maximize aggregate household utility, as opposed to a collective household model.

fertility in the previous period, $\pi_{t-1}n_{t-1}$, and per adult labor supply ℓ_t :

$$L_t = \pi_{t-1}n_{t-1}\ell_t. \quad (4)$$

2.2 Equilibrium

Labor markets clear so that

$$L_t = L_{at} + L_{mt}.$$

Wages are pinned down by the manufacturing firm's marginal product

$$w_{mt} = p_{mt}A_{mt}h_t$$

and equation (3). Given firm profit maximization, the equilibrium agricultural employment share is therefore

$$\frac{L_{at}}{L_t} = \left(\frac{\theta p_{at}A_{at}}{\xi p_{mt}A_{mt}h_t} \right)^{\frac{1}{1-\theta}} \frac{T}{L_t}. \quad (5)$$

The land-labor ratio, $\frac{T}{L_t}$, captures the Malthusian land congestion mechanism: the higher the labor force relative to land, the lower the fraction of labor will work in agriculture. Our second key mechanism is that increases in human capital, h_t , will reduce employment share in agriculture as the manufacturing wage rises. Finally, the greater are distortions (smaller ξ), the more employment is kept inefficiently in agriculture.

Each household's optimal choice of fertility and child education are as follows:

$$n_t = \frac{(\alpha - \eta\gamma)\mu\bar{w}_t}{(1 + \delta + \alpha + \sigma)(\mu\pi_t\bar{w}_t\phi - p_{ft})}, \quad (6)$$

$$e_{t+1} = \frac{\eta\gamma(\mu\pi_t\bar{w}_t\phi - p_{ft})}{(\alpha - \eta\gamma)\mu\pi_t\bar{w}_t}. \quad (7)$$

To ensure a non-negative solution, we assume that $\mu\pi_t\bar{w}_t\phi > p_{ft}$, which ensures that the expected time-cost of a child exceeds the cost of preventing conception. We extend the model in Online Appendix A.2 to allow households to incorporate general equilibrium wage feedback into their decisions; doing so does not change our core results.

To characterize how demographic transition affects household living standards, we consider per-capita expenditure on consumption goods excluding contraception. Define

$$E_t \equiv \frac{p_{at}c_{at} + p_{mt}c_{mt}}{n_{t-1} + \pi_t n_t}.$$

Given household utility maximization, per-capita expenditure is then

$$E_t = \frac{1 + \delta}{1 + \delta + \alpha + \sigma} \left[1 - (1 - \xi) \frac{L_{at}}{L_t} \right] \frac{w_{mt}}{n_{t-1} + \pi_t n_t}. \quad (8)$$

2.3 Effect of Changes in Family Planning and Child Survival

We assess the effect of demographic transition on sectoral employment and per capita consumption through the lens of our model by considering (i) a reduction in the price of the family planning technology p_{ft} , and (ii) an increase in child survival π_t . The price p_{ft} includes both monetary and non-monetary costs associated with accessing the family planning technology.

2.3.1 Effects on Fertility and Education

As seen in equations (6) and (7), reducing p_{ft} decreases fertility and increases education of the next generation:

$$\frac{\partial n_t}{\partial p_{ft}} > 0, \quad \frac{\partial e_{t+1}}{\partial p_{ft}} < 0. \quad (9)$$

That is, cheaper family planning reduces fertility and raises education per child due to the quantity-quality tradeoff.

We also characterize the effect of improved child survival. Higher π_t reduces births and raises education per child:

$$\frac{\partial n_t}{\partial \pi_t} < 0, \quad \frac{\partial e_{t+1}}{\partial \pi_t} > 0. \quad (10)$$

With higher chances of survival, the expected cost of a birth rises, driving fertility down. Turning to education, each birth is now more expensive in expectation given the higher probability the infant survives. Since couples enjoy sex, they increase expenditure on family planning, reducing their net income. With fewer births and hence a lower childrearing time cost $\phi \pi_t n_t$, households devote more time to educating their remaining children.

Net fertility $N_t = \pi_t n_t$ moves with gross fertility:

$$\frac{\partial N_t}{\partial p_{ft}} > 0, \quad \frac{\partial N_t}{\partial \pi_t} < 0.$$

Although higher π_t mechanically raises the number of surviving children for a given number of births, it also raises the expected time cost of each birth. Households respond by reducing births more than proportionally and by investing more in each surviving child.

These comparative statics result in the following empirical prediction:

PREDICTION 1: *A reduction in p_{ft} and an increase in π_t unambiguously reduce fertility and net fertility, and raise education.*

Proof: see Online Appendix [A.3](#).

2.3.2 Contemporaneous Effects on Agricultural Employment Share

In the short run, both current-generation human capital h_t and current-generation adult population $\pi_{t-1}n_{t-1}$ are predetermined. The only contemporaneous variable shaping sectoral allocations that may change is per-adult labor hours, ℓ_t . A reduction in p_{ft} reduces fertility and increases education per child. In our model, these two margins exactly offset in total education time because $e_{t+1}\pi_t n_t$ is constant, so changes in ℓ_t are driven solely by the childrearing time cost $\phi\pi_t n_t$. As a result, lower p_{ft} and higher π_t increase per-adult labor supply in equilibrium. Empirically, however, these labor-supply responses are often modest, especially in less developed contexts ([Aaronson et al. 2021](#); [Bleakley 2007](#)). While women may reduce their labor supply in response to fertility ([Bloom et al. 2009](#)), men may increase theirs ([Lundberg and Rose 2002](#)), potentially offsetting the household-level impact of changes in fertility.

2.3.3 Next-Generation Effects

In subsequent generations, more accessible family planning and higher child survival have two additional effects. First, human capital (h_{t+1}) rises, thereby pulling workers into the manufacturing sector when human capital is useful. Second, the adult population ($\pi_t n_t$) falls, reducing land congestion in agriculture.

$$\frac{\partial \ln(L_{at+1}/L_{t+1})}{\partial p_{ft}} = -\frac{1}{\ell_{t+1}} \frac{\partial \ell_{t+1}}{\partial p_{ft}} - \frac{\eta}{1-\theta} \frac{1}{e_{t+1}} \frac{\partial e_{t+1}}{\partial p_{ft}} - \frac{1}{n_t} \frac{\partial n_t}{\partial p_{ft}}, \quad (11)$$

$$\frac{\partial \ln(L_{at+1}/L_{t+1})}{\partial \pi_t} = -\frac{1}{\ell_{t+1}} \frac{\partial \ell_{t+1}}{\partial \pi_t} - \frac{\eta}{1-\theta} \frac{1}{e_{t+1}} \frac{\partial e_{t+1}}{\partial \pi_t} - \frac{1}{n_t} \frac{\partial n_t}{\partial \pi_t} - \frac{1}{\pi_t}. \quad (12)$$

Relative to the prior period in which only per-adult labor supply ℓ_t may change, total population now also changes. The first term in equations (11) and (12) captures the effect of per-adult labor supply, which we discussed above in Section 2.3.2. The second term captures the role of human capital in next-generation structural transformation. The final terms of equations (11) and (12) capture the effect of land congestion. The population term in both equations, $-\frac{1}{n_t} \frac{\partial n_t}{\partial x_t}$, captures adjustment on the margin of fertility. Land congestion has an extra term, $-\frac{1}{\pi_t}$, in equation (12) to account for the direct effect of child survival increasing the number of adults next period.

Thus, rising human capital reduces the agricultural employment share, while reduced land congestion raises it. Which force dominates depends on their relative strength, yielding the following prediction:

PREDICTION 2: *A reduction in p_{ft} and an increase in π_t :*

- (i) *raises agricultural employment share if $\eta < 1 - \theta$,*
- (ii) *reduces agricultural employment share otherwise.*

Proof: see Online Appendix [A.3](#).

In sum, the theoretical net effect of demographic transition on structural transformation is ambiguous, depending on just two parameters: η and θ . The model generates sharp predictions regarding mechanisms: holding other factors constant, reduced population pressure slows down structural change through land congestion, while increased human capital accelerates it. We use a unique quasi-experiment in Bangladesh to empirically test which mechanism dominates in an open economy setting.

With η being the elasticity of human capital with respect to education, and $-1/(1-\theta)$ the elasticity of agricultural employment share with respect to human capital, $\eta < 1 - \theta$ implies that diminishing returns are strong enough (low η) or agriculture is land-intensive enough (high $1 - \theta$) that the implied increase in effective labor is too small to offset the relaxation of land scarcity. [Boppart et al. \(2023\)](#) estimate the average agricultural land share among poor countries to be about 0.2, with Bangladesh exhibiting one of the highest at 0.36. Returns to education in developing countries are often quite low due to poor quality education and few opportunities to apply advanced skills, suggesting η may be low in Bangladesh.⁸

Demographic transition may also affect household living standards through the same mechanisms that shape structural transformation. As shown in equation (8), per-capita expenditure responds to changes in household size, human capital, and sectoral employment. Our final prediction concerns how per-capita expenditure responds.

PREDICTION 3: *A reduction in p_{ft} and an increase in π_t have an ambiguous effect on E_t :*

- (i) *the associated reduction in N_t and increase in w_{mt} raise E_t ;*
- (ii) *an associated increase in L_{at}/L_t lowers E_t .*

Proof: see Online Appendix [A.3](#).

Prediction 3 highlights the ambiguous effects of a simultaneous reduction in net fertility and increase in the agricultural employment share. On the one hand, fewer mouths to

⁸[Schoellman \(2012\)](#), for example, finds Bangladeshi immigrants to the U.S. have the 6th lowest returns to origin education among all origin countries.

feed within the household raises per-capita expenditure. Moreover, the quantity-quality tradeoff yields higher wages in the manufacturing sector, and hence greater income and more consumption per capita. On the other hand, a smaller population leads to a greater share of wages coming from the lower-paid agricultural sector, which reduces per-capita consumption.

The model applies most directly to economies open to trade. With sufficiently high trade costs, demand-side forces may reverse the population size prediction because a larger population raises demand for local food production. We show in Online Appendix A.4 that our results hold even with trade frictions, so long as trade costs are not too large.⁹ Consistent with [Cavalcanti et al. \(2021\)](#), we expect general equilibrium price effects of family planning and child health programs to be modest in our setting.

3 Quasi-Experimental Household-Level Evidence

We examine the model’s Predictions 1–3 using the Maternal and Child Health and Family Planning (MCH-FP) program in Matlab, Bangladesh (1977–1988). The MCH-FP provided family planning services and child health interventions to treated villages but not to comparison villages.

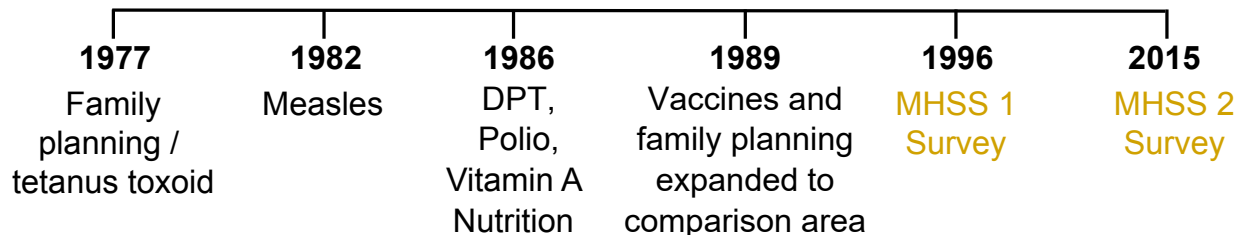
3.1 The Intervention

The MCH-FP program was implemented by icddr,b (formerly known as the International Centre for Diarrhoeal Disease Research, Bangladesh) in Matlab, Bangladesh, in two distinct phases. Phase 1 (October 1977–December 1981) provided modern contraceptives and maternal health services through home visits ([Bhatia et al. 1980](#); [Phillips et al. 1982](#)). Phase 2 (starting in 1982) added intensive child health interventions including vaccines ([Fauveau 1994](#)). The staggered program rollout led to differential exposure by birth cohort. Many interventions were expanded to the comparison area in 1989, creating an experimental period from 1977 to 1988. Figure 1 shows the timeline of the program and subsequent survey data collection.

The MCH-FP program was introduced using a quasi-experimental design, with 70 out of 149 villages assigned to receive the intervention and the remaining villages to an untreated comparison group (see Appendix Figure C.1). Villages were grouped into four contiguous treatment blocks and two comparison blocks based on distance to health clinics and

⁹A growing literature emphasizes an open-economy perspective on structural change ([Uy et al. 2013](#); [Sposi 2019](#); [Fajgelbaum and Redding 2022](#); [Farrokhi and Pellegrina 2023](#); [Gollin et al. 2025](#)).

Figure 1: Timeline of Mother and Child Health and Family Planning Program and Follow-up Surveys



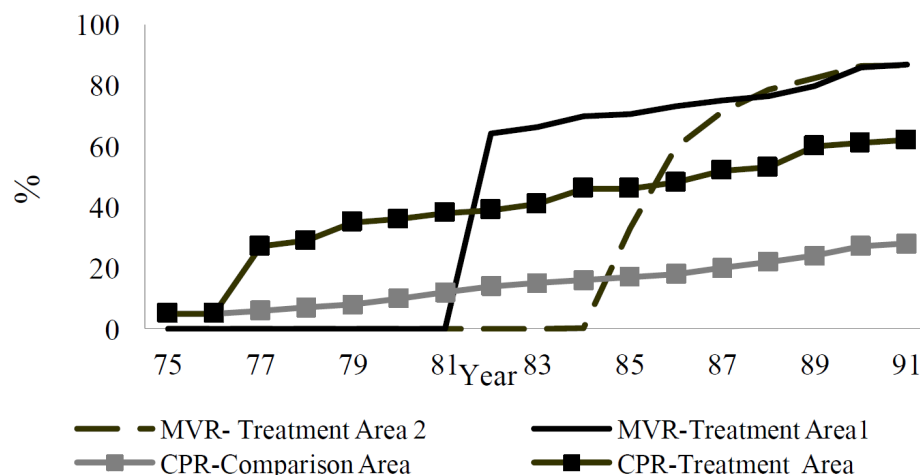
to balance pre-intervention socioeconomic characteristics. This design facilitated program implementation while minimizing information and vaccine spillovers (Huber and Khan 1979).

A key component of the program was home visits by female health workers. In Matlab, purdah—the practice of female seclusion—is common and substantially constrained women’s access to medical care. Because government family planning services in the comparison area were delivered primarily through clinics, the MCH-FP’s home visits sharply reduced the non-monetary cost of obtaining contraception in treatment villages. In the context of our theoretical model, we therefore consider the program as primarily reducing p_{ft} , the cost of contraception.

Health workers provided a variety of family planning methods in the homes of the beneficiaries including condoms, oral pills, vaginal foam tablets, and injectables. In addition, beneficiaries were informed about fertility control services provided by the project in health clinics such as intrauterine device insertion, tubectomy, and menstrual regulation. During these visits female health workers also provided counseling on contraception, nutrition, hygiene, and breastfeeding, and encouraged women to continue using contraceptives. These services were supported by followup and referral systems to manage side effects and continued use of contraceptives (Phillips et al. 1982; Fauveau 1994).

The program succeeded in changing its targeted outcomes: fertility and vaccination rates. Prior to the program, the contraceptive prevalence rate (CPR) for married women age 15–49 was low at less than 6 percent in both the treatment and comparison areas. The CPR reached 30 percent in the treatment area in the first year, then rose steadily, reaching almost 50 percent by 1988, as depicted in Figure 2. Because contraceptives were also provided by the government, the CPR increased in the comparison area, but not as quickly, and remained below 20 percent in 1988. By 1990, there was still a 20 percentage point difference in the

Figure 2: Trends in contraceptive prevalence rate (CPR) and measles vaccination rates (MVR) for children 12-59 months by calendar year



Source: Replicated from Figure 2 in [Barham et al. \(2023\)](#).

CPR between the two areas.

During the program's second phase, which started in 1982, intensive child health interventions were rolled out including vaccination against measles, tetanus, pertussis, polio, and tuberculosis, as well as nutrition rehabilitation. The measles vaccination rate surged to 60 percent in 1982 after it was introduced in half of the treatment area, and in 1985 when it was introduced in the other half as shown in Figure 2. By 1988, coverage rates for children aged 12–23 months living in the treatment area were 93 percent for the vaccine against tuberculosis, 83 percent for all three doses of the vaccines against diphtheria, pertussis, tetanus, and polio, 88 percent for measles, and 77 percent across all three major immunizations ([icddr, b 2007](#)). Government services did not regularly provide measles vaccination for children until around 1989, so the comparison area was an almost entirely unvaccinated population ([Koenig et al. 1991](#)). Nationally, the measles vaccination rate for children under the age of five was less than 2 percent in 1986 ([Khan 1998](#)) and below 40 percent in the comparison area in 1990 ([Fauveau 1994](#)).

Together, the family-planning and child-health components affected both margins determining net fertility: births and child survival. Using MHSS2 pregnancy histories, [Barham et al. \(2021\)](#) estimate that the MCH-FP reduced woman-level gross fertility—measured by children ever born—by about 0.6 births, a roughly 10 percent decline. Improved child survival partly offset this decline: the program reduced child deaths by approximately 0.2 per woman, leaving a reduction in net fertility of approximately 0.4 surviving child, or 7 percent. Thus, the program generated the demographic changes noted in the model's Prediction 1: lower gross fertility, higher child survival, and lower net fertility.

3.2 Data and Analysis Sample

Data Sources. We draw on the exceptionally rich data available for the Matlab study area. Our primary source is the Matlab Health and Socioeconomic Survey (MHSS), with the 1996 wave (MHSS1) ([Rahman et al. 1999](#)) providing medium-term and the 2012–2014 wave (MHSS2) ([Barham et al. 2014](#)) providing longer-term followup data.

MHSS2 tracked MHSS1 respondents, their children, and those who left Matlab prior to 1996. Fieldwork occurred across multiple years with increasing effort in order to maximize response rates among difficult-to-track migrants. Migrants were identified as a part of the survey and tracked throughout the country. Beginning in October 2013, rapid-response teams were put in place in major city centers in Bangladesh so that interviews could take place once a migrant was found via family members in Matlab. In-person surveys were collected during the two Eid festivals in July and October 2014 when migrants returned to their villages in Matlab. Fifteen percent of men in our sample, international migrants living abroad, were contacted using a phone survey.

We use two supplementary data sources. First, two censuses in 1974 and 1982 ([icddr, 1974, 1982](#)) provide information on pre-program household characteristics. Second, the Matlab demographic surveillance system (DSS) data captures all vital events, i.e., births, marriages, deaths, in and out migration events, between 1974 and 2014 for all individuals living in the Matlab study site. The DSS was crucial for tracking individuals who left Matlab. A key feature of all these data is that individuals can be linked across different data sources over time by a unique individual identifier. There are few, if any, other study sites that have similarly rich data availability to allow for this type of long-term evaluation.

Household aggregation. Household size may endogenously shape future household splits ([Foster and Rosenzweig 2002](#)). To address this issue, we fix the composition of households as of 1996, before program-impacted cohorts are old enough to begin leaving their childhood households.

Hence, the MHSS1 household serves as our unit of analysis. We follow all household members—including migrants not interviewed at the time of MHSS1—and their descendants to the MHSS2 survey, and aggregate MHSS2 individuals to the MHSS1 household they link back to.

Our MHSS2 sample includes individuals who resided in or descended from an MHSS1 household. Consequently, every MHSS2 respondent in our sample links back to a single MHSS1 household, making it relatively straightforward to aggregate individual-level outcomes to the MHSS1 household level.

Intent-to-Treat Assignment. Access to the MCH-FP program was determined by a household’s village of residence during the program period. However, we do not use location at the time of MHSS surveys, or birthplace for individuals born after the program began, because residential location may have been affected by the program (Barham and Kuhn 2014).

Treatment assignment is instead based on 1974 census village of residence, or pre-1977 village if not found in the census, to avoid endogenous sorting. We trace each individual to their 1974 household, or if not yet born, their first DSS-linked household head. For an individual, the ITT variable takes the value of 1 if the 1974 census-linked individual or traced-back household head was living in a village in the treatment area in the 1974 census or first migrated into a village in the treatment area from outside Matlab between 1974 and 1977, and 0 otherwise. For our primary household-level analysis, the treatment indicator for the MHSS1 household is derived from the individual treatment indicator of the household head.

Analysis Sample. MHSS1 was a seven percent random sample survey of household compounds, called *baris*, in the Matlab area. In each *bari*, two households were selected for interview: a primary household selected randomly, and a secondary household selected purposively. Within a household, individuals (aged 6 or older) were randomly sampled for in-depth interviews.

We begin by building a sample of households using the primary households that were randomly selected for interview in MHSS1. We select households where the household head also was chosen for an individual interview in the MHSS1 Household survey. From this set, we remove households where the household head could not be linked back to a treatment status or who could not be linked back to the Matlab area (i.e., the DSS) prior to the start of the MCH-FP in October 1977. These criteria result in a set of 2,534 households. When measuring individual-level outcomes in MHSS1, we consider outcomes from working-age survey respondents from these households.

To assess the role of human capital as a mechanism, as done in Section 4.2, we analyze employment outcomes at the individual level. This analysis relies on the same set of individuals that underlie our household-level analysis. Because we are focused on employment outcomes, we additionally restrict the individual-level analysis to individuals born in 1947 or later, and were thus 65 or younger at the start of MHSS2 surveying, and to those who were born in 1988 or earlier, to focus on the set of individuals born before or during the MCH-FP evaluation period.

Due to low attrition in MHSS2, fewer than 2 percent of MHSS1 households have no members who were tracked to the MHSS2 survey round. Our sample restrictions and attrition

result in a MHSS1 household-level sample using MHSS2 data of 2,484. At the individual level, including death and other types of non-response, the attrition rate is 16 percent among all men in our sample, and 11 percent among men born during the program interventions. This is a low attrition rate despite 60 percent of men in our sample migrating out of the Matlab study area, 25 percent of whom migrated internationally. Our individual-level sample consists of 4,744 men and 4,628 women.

3.3 Empirical Strategy

3.3.1 Pre-Program Balance and Trends

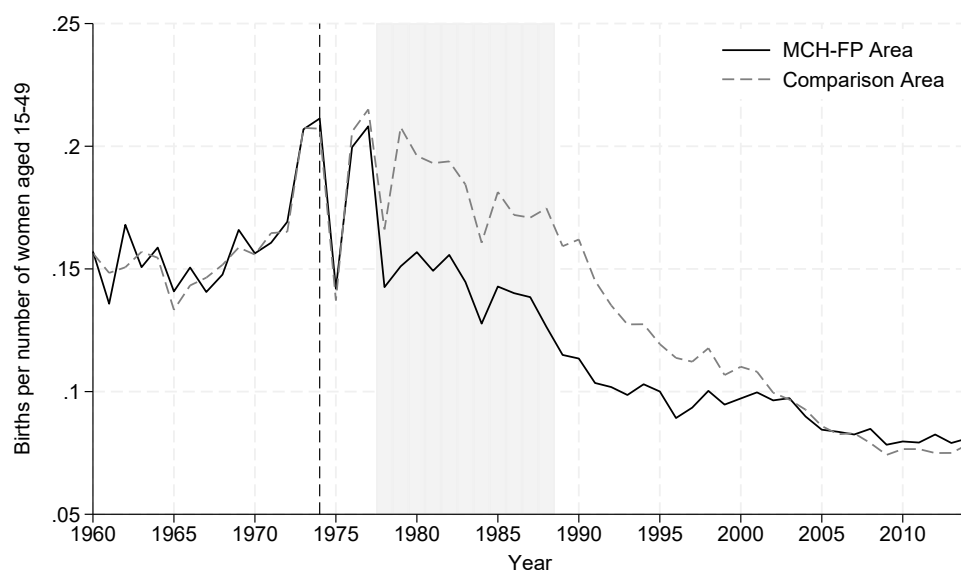
Treatment and comparison areas were designed to be socially and economically similar (Phillips et al. 1982), and prior studies extensively document pre-intervention balance (Koenig et al. 1990; Menken and Phillips 1990; Barham 2012; Joshi and Schultz 2013; Barham and Kuhn 2014; Barham et al. 2023).

We further demonstrate that the demographic transition was similar between the treatment and comparison areas prior to the program in Figure 3. Birth rates were nearly identical in levels and trend in the two decades prior to the program, including the drop in 1975 due to the Bangladesh famine. During the program period (1977–1988), shaded in gray, birth rates diverged substantially: the comparison area experienced nearly 30 percent higher birth rates than the treatment area, consistent with Prediction 1 of the model.

Additionally, migration stocks and flows were similar between the treatment and comparison area at the start of the program and through to 1982, for the cohort of individuals most likely to migrate at the start of the program (Barham and Kuhn 2014). Barham et al. (2023) further show that men born between 1977 and 1988 come from households experiencing similar labor market outcomes in 1974, 1982, and 1996. Finally, Barham (2012) shows that cognitive functioning, height, and years of education were similar across the treatment and comparison areas in 1996 for those who were old enough that their human capital and height were not likely to have been affected by the program.

We demonstrate that this balance persists in our analysis sample of MHSS1 households in Appendix Table C.1, complementing previous work examining individual differences among the panel of MHSS1 respondents. Column 7 presents normalized differences—the difference divided by the comparison area standard deviation—to provide an indication of the economic significance of the differences independent of sample size. Normalized differences bigger than 0.25 standard deviations are generally considered to be economically meaningful (Imbens and Wooldridge 2009). Similar to the previous literature, this analysis shows that tubewell water and religion are unbalanced.

Figure 3: Trends in Birth Rate 1960–2014, MCH-FP Treatment Area and Comparison Area



Notes: The figure reports birth rate estimates from the Matlab Demographic Surveillance System (DSS) between 1960 and 2014. DSS records begin April 30, 1974 (denoted by the vertical dashed line). In 1974 and earlier, we construct birth rates by counting the number of individuals who resided in the DSS area on April 30, 1974 who were born in a given year and dividing by the number of women aged 15–49 residing in the DSS on April 30, 1974. In later years, birth rates are constructed by counting the number of individuals residing in the DSS on the day of their birth in a given year and dividing by the number of women aged 15–49 residing in the DSS area on January 1 of that year. The shaded gray area marks the period (1977–1988) when the MCH-FP intervention was available in the treatment area, but not the comparison area.

The difference in tubewell water access is 0.43 standard deviations. [Barham \(2012\)](#) finds no evidence that tubewell access biases medium-run estimates of the program’s effect on human capital. In addition, as a robustness check on our baseline estimates, we examine effects for the Muslim population only and find results are similar. We control for all baseline characteristics, including tubewell water access and household religion, in all regressions.

One further concern is that individual and household attrition between MHSS1 and MHSS2 may be unbalanced by treatment status. We do not find evidence for this concern. Nevertheless, to account for the small amount of attrition we observe, we construct inverse propensity weights that predict household-level attrition using the set of baseline characteristics reported in Appendix Table [C.1](#) as well as their interaction with treatment assignment.

Similarly, in our individual-level analysis we construct weights to account for attrition in MHSS2 among our individual panel sample following [Barham et al. \(2023\)](#). Appendix Table [C.2](#) reports comparison area attrition rates for men (16.3 percent) and women (11.8 percent) in the individual sample, and tests for differential selection into non-response between the treatment and comparison area. We fail to reject the null that response rates are the same across areas, and while some individual characteristics do show differential associations with attrition, we do not find strong joint evidence of systematic differential selection.

3.3.2 Empirical Specification

To examine the effect of the MCH-FP program on sectoral employment and agricultural outcomes, we take advantage of well-balanced treatment and comparison areas and use a single difference intent-to-treat (ITT) model. We estimate the household-level specification,

$$Y_h = \omega_0 + \omega_1 T_h + \zeta X_h + \varepsilon_h, \quad (13)$$

where T_h is an indicator for whether household h is treated, X_h is the vector of demographic and baseline characteristics listed in Appendix Table [C.1](#), and Y_h is a sectoral employment or expenditure outcome. The coefficient of interest, ω_1 , measures the difference in average outcomes between treatment and comparison area households conditional on the set of 1974 household characteristics. To adjust our inference for the village-level treatment assignment, we cluster standard errors by the village of the household head of h or his antecedents, traced back following our ITT assignment. We also show our main results are robust to alternative methods of inference in Appendix Table [C.4](#), discussed in Section [3.4](#).

Our identification assumption is that average post-program outcomes among treatment area households, conditional on pre-intervention controls, would have been the same as the

average outcomes in the comparison area, had the treated not been treated. While this is not a testable assumption, the assumption seems plausible given the baseline balance in levels and trends in birthrates as well as many other household characteristics discussed Section 3.3.1.

3.4 Effects on Sectoral Employment

Table 1 presents ITT estimates of the MCH-FP program’s effects on household-level sectoral labor allocation. Employment outcomes are defined over household members age 15 and up. We differentiate medium-run impacts, measured 19 years after program initiation using the 1996 MHSS1 survey, from long-run impacts observed 35 years after in the 2012–2014 MHSS2 survey. As most cohorts exposed to the MCH-FP had not yet entered the labor market in the medium-run, we anticipate only modest effects operating through parental labor supply (Section 2.3.2). The 35-year horizon using MHSS2 additionally enables examination of the land-congestion and human-capital mechanisms outlined in Section 2.3.3.

Outcome variable definitions differ slightly across surveys. In MHSS1, the outcomes capture the share of months per year that household members worked in agriculture or non-agriculture (columns 1 and 2).¹⁰ In MHSS2, the dependent variables are the share of total annual hours worked in each sector (columns 3–5),¹¹ as well as the average total number of annual hours worked by household members (column 6).¹²

In the medium-run, effects on sectoral shares are close to zero and not statistically significant, in line with the fact that treated cohorts had not yet entered the labor market.

In the long run, once treated cohorts entered the labor force, the effects are substantial and consistent with Malthusian land congestion dominating human capital channels. The MCH-FP increased adults’ agricultural work share by 4.1 percentage points (20 percent of comparison households’ mean; column 3) and reduced manufacturing work share by 3.2 percentage points (16 percent of comparison households’ mean; column 4). Both effects are statistically significant at the 5 percent level. In contrast, the effect on share of workers in services is small (-3 percent; column 5) and not statistically significant. The effect on average annual hours worked per person is negligible (-2 percent; column 6) and not statistically significant, suggesting that the changes in sectoral time allocations are not driven by shifts

¹⁰Because work time is measured in months, individuals working in both sectors within a single month may cause shares to sum to more than one.

¹¹Employment shares may not sum to one for two reasons: (i) results for the construction sector are not reported, and (ii) households with no reported work are coded as allocating 0 percent of time to each sector.

¹²We exclude analyzing the effect on total months worked in MHSS1 because of a lack of variation at this extensive margin with most adults working either 0 or 12 months per year in our data. There is no data on the intensive margin of hours worked in MHSS1.

Table 1: ITT Effects of MCH-FP on Work Time Shares by Sector: Household-Level

	MHSS1 (1996)		MHSS2 (2012–2014)			
	(1) Agriculture Share	(2) Non-Agriculture Share	(3) Agriculture Share	(4) Manufacturing Share	(5) Services Share	(6) Annual Hours Per Person
Treatment	0.007 (0.021)	0.004 (0.021)	0.041*** (0.014)	-0.032** (0.014)	-0.013 (0.018)	-27.083 (35.457)
% chg. rel. to mean	1.1	1.2	19.9	-15.8	-2.8	-1.9
Mean	0.68	0.36	0.21	0.20	0.48	1445.47
Baseline controls	Y	Y	Y	Y	Y	Y
Observations	2534	2534	2484	2484	2484	2484

Notes: The table presents estimates of equation (13) for outcomes at the MHSS1 household-level. Variable means refer to the comparison group. Standard errors are clustered by the 1996 household head’s pre-program village. Columns (1) and (2) measure outcomes in the 1996 MHSS1, while columns (3) through (6) measure outcomes in the 2012–2014 MHSS2. The MHSS1 dependent variable is the share of working months in the year allocated to each sector. MHSS2 dependent variables are the share of hours worked by sector within the household (columns 3–5) and the average annual hours worked per person aged 15 or older (column 6). Employment hour shares do not sum to 1 for two reasons. First, we do not report results for the construction sector. Second, we code the small set of households who do not work as spending 0 percent of their time working in each sector. MHSS2 regressions are weighted to account for household-level attrition between the MHSS1 and MHSS2 surveys. Due to changes between survey waves, sectors are constructed differently in the MHSS1 and MHSS2, and therefore are not directly comparable. See Appendix B for more details on how we classify workers into sectors. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

in total hours worked.

These findings are consistent with returns to education being strongly diminishing (a low η) and agriculture being sufficiently land intensive (a high $1 - \theta$), such that $\eta < 1 - \theta$, as in the model’s Prediction 2(i).

To assess the depth and breadth of the structural transformation impacts of the program, we look at several related outcomes.

Given the importance of entrepreneurship for development (McMillan and Woodruff 2003; Buera et al. 2011, 2021), Appendix Table C.3 examines whether the patterns observed in employment are matched by sector-specific entrepreneurship, defined as someone in the family owning a business in the sector. Columns 1 through 3 show a similar pattern as Table 1: the MCH-FP increased agricultural entrepreneurship by 20 percent (significant at the 1 percent level), with no change in manufacturing or services entrepreneurship.

Since large firms, especially factories, drive structural change and growth (Buera and Kaboski 2012), we further examine how the MCH-FP affected employment across firm types in Appendix Table C.3. Employment at factories among treated households lagged behind comparison area households by 14-23 percent (significant at the 5 percent level or below; columns 4 and 5), as did employment at large firms by 21 percent (significant at the 5 percent level; column 6).

Taken together, our results indicate that the MCH-FP program slowed the pace of structural transformation across an array of outcome measures. Interpreted through the lens of the model from Section 2, this pattern is consistent with the model’s Prediction 2(i) ($\eta < 1 - \theta$) that the Malthusian land congestion mechanism dominates both the quantity-quality tradeoff and any human capital gains driven by the child health interventions.

Inference. We use village-level clustering as our baseline. Treatment eligibility was operationally determined at the village level, and village clustering is consistent with the approach used in most previous evaluations of the Matlab experiment. Nevertheless, we test the robustness of our baseline results to alternative inference approaches, with p-values reported in Appendix Table C.4. The treatment coefficient for agricultural share always remains statistically significant, while the manufacturing share coefficient is more sensitive.

One concern may be that the Matlab study site has just six geographic blocks for assigned treatment: four were treated, two were not. So few treatment blocks may mean that our baseline standard errors, clustered at the village level, are insufficiently conservative. Clustering at the block level yields p-values of 0.024 for agricultural employment share and 0.048 for manufacturing. With only six blocks, however, conventional cluster asymptotics may be unreliable (Cameron et al. 2008). We therefore compute a wild-cluster bootstrap using Webb’s (2023) six-point weights, yielding p-values of 0.048 for agriculture and 0.093 for manufacturing.

We also conduct randomization inference by constructing 10,000 placebo village treatment assignments that preserve the geographic contiguity and compactness of the actual treatment area. This is the most conservative procedure for our main outcome, yielding a p-value of 0.066 for agriculture and 0.330 for manufacturing. We do not, however, view these permutations as necessarily reproducing the original assignment mechanism. Block formation reflected not only compactness and contiguity but also proximity to pre-existing health facilities, so that households could access the health services accompanying the intervention; we do not observe the historic locations of these facilities. More fundamentally, the original assignment protocol cannot now be precisely reconstructed from surviving records, and nearly all individuals responsible for the MCH-FP’s implementation have passed away.

Finally, another concern is that we estimate the treatment’s effect multiple outcomes. We therefore adjust for multiple hypothesis testing across the outcomes in Table 1 using the false-discovery-rate procedure of Anderson (2008) produces adjusted p-values of 0.019 for agriculture and 0.072 for manufacturing.

Robustness. Our baseline results are broadly robust to a range of alternative sample choices and specifications.

One concern is that areas closer to treatment and control borders differ due to spillovers (informational or from vaccines) or because the areas could be less similar since they are farther apart. To explore this possibility we examine effects for the sample living in a village prior to the intervention with a centroid within 3km of the treatment-control border, and entirely excluding border villages. Findings in panels B and C of Appendix Table C.5 demonstrate that magnitudes are similar to our baseline results from Table 1.

Given the finding from Appendix Table C.1 that Muslims are disproportionately represented in comparison villages, we re-estimate equation (13) using only Muslim households in Panel D of Appendix Table C.5. Again, results are virtually unchanged. Since Matlab is about 85% Muslim, we do not have sufficient statistical power to estimate program effects for the Hindu population on its own.

We address another asymmetry between treatment and control areas, which is that the only urban center in Matlab is located in the treatment area. Panel E of Appendix Table C.5 shows our baseline estimates when excluding this urban center from the sample. The results are the same in sign and moderately stronger in magnitude and precision.

We additionally examine robustness of our baseline results to the construction of a flood embankment. Built in 1987, the embankment may have changed the returns to agriculture by protecting some land and exposing others to flooding. Appendix Table C.6 presents two robustness checks that address whether the embankment drives our results. Panel A displays our baseline results. Panel B shows effects on our baseline outcomes when controlling for whether a village was completely eroded by the river as well as the log distance from households' traceback village centroid to both the embankment and the river. The impact of the MCH-FP on treatment remains strongly positive and statistically significant, while the point estimate for manufacturing falls in magnitude and loses statistical significance. Panel C shows the impact of the MCH-FP stratified by whether the household's traceback village was protected by the embankment or not, still controlling for log distance to the embankment and river. Consistent with the embankment increasing the returns to agriculture, protected villages experience a stronger agricultural employment share response to the MCH-FP. Yet we still see a positive and statistically significant effect of the MCH-FP on agricultural employment share even in unprotected villages.

Finally, we conduct robustness related to sample attrition. Appendix Table C.7 provides attrition bounds by either imputing missing values based on the worst-case scenario (i.e., minimum and maximum values of outcome) or by imputing values at the comparison/treatment area mean $\pm$ 0.1 standard deviation following the approach of Kling and

[Liebman \(2004\)](#). Treatment coefficients for agriculture share are always positive, and for manufacturing always negative, consistent with our baseline. When Kling-Liebman bounds are applied, the coefficients also remain statistically significant at the 5% level.

Rural-to-Urban Migration. To understand how sectoral employment is reallocated, we examine rural-to-urban migration in our sample. Rural-to-urban migration is central to structural transformation ([Lagakos 2020](#); [Lagakos et al. 2023](#)), as it enables workers to move between sectors. In Bangladesh, the bulk of growth in manufacturing employment occurs in cities. As shown in Appendix Table [C.8](#), baseline sectoral employment effects are driven by reduced urban employment and increased rural agricultural work. Treated households work 3.7 percentage points more in rural agriculture, a 16% increase. Treated households work 3.3 percentage points less in urban manufacturing relative to comparison households, a 24% reduction. Both estimates are statistically significantly different from zero. Hence, reduced population pressure slows structural transformation by reducing movement from rural agriculture into urban industry, rather than by encouraging rural industrialization.

3.5 Sectoral Consumption

We next examine program impacts on expenditure, the subject of the model’s Prediction 3. To do so, we construct sectoral consumption measures at the household level. The MHSS2 asks households about expenditure on food in the week prior to the survey, and expenditure on non-food goods and services over the month or 12 months prior to the survey. We assign each item to a sector based on COICOP ([United Nations 2018](#)).

First, we test the model’s Prediction 3. As human capital rises and the number of children decreases, per capita consumption should increase. Conversely, a higher share of household work time in agriculture should reduce per capita consumption to the extent that agriculture pays inefficiently low wages. Panel A resolves the ex-ante ambiguous sign of the consumption response: the MCH-FP increased per-capita expenditure. Overall, the effect is an 11% increase in per capita expenditure, as seen in column 4.

We next test whether expenditure shares changed as a result of the MCH-FP. The open-economy assumption in our model implies that sectoral change cannot be driven by shifts in demand.

We find modest impacts of the MCH-FP on sectoral spending shares, with results reported in Panel B of Table [2](#). Households allocate a lower share of expenditure to agriculture and a greater share to services, with only the coefficient for services being statistically significantly greater than zero. Importantly, the direction of the effect goes in the opposite direction

Table 2: ITT Effects of Consumption Shares by Sector

	(1)	(2)	(3)	(4)
	Agriculture	Manufacturing	Services	Total
<i>Panel A: Per capita in USD</i>				
Treatment	15.81*	4.28	29.30**	49.39***
	(9.40)	(3.53)	(11.79)	(18.21)
% chg. rel. to mean	6.9	4.8	21.1	10.8
Mean	228.36	88.60	138.57	455.52
Observations	2013	2013	2013	2013
<i>Panel B: Share of total consumption</i>				
Treatment	-0.012	-0.004	0.016*	
	(0.008)	(0.004)	(0.008)	
% chg. rel. to mean	-2.2	-2.2	6.0	
Mean	0.536	0.200	0.264	
Observations	2013	2013	2013	

Notes: The table presents estimates of the effect of the MCH-FP on 2014 consumption outcomes summed to the MHSS1 household-level. The sample is restricted to MHSS1 households where MHSS2 consumption was observed within at least one household. Consumption goods classified into sectors based on COICOP (United Nations 2018). Panel A divides consumption expenditure per sector by the number of rostered individuals present in the household. The outcome in Panel B is the share of total consumption within each sector. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

of employment: treated households work more in agriculture but spend less on the sector, suggesting that the employment-based structural transformation is not demand-driven.

4 Mechanisms and Adjustment

Using the rich household- and individual-level data, we examine the mechanisms highlighted by our theoretical model, land congestion and human capital. We then provide complementary evidence on how households adjust their agricultural practices in response to the program.

4.1 Land Congestion

We first examine whether the long-run program effects are consistent with a land congestion channel, whereby smaller cohorts raise the agricultural work share. At the local level land is fixed. But at the household level, families may adjust to changes in household size by buying or selling land. In Appendix Table C.9, we find no evidence that the program affected the

Table 3: ITT Effects of MCH-FP on Household Size and Composition

	(1) Number of Men Age 24–34	(2) Number of Women Age 24–34
Treatment	-0.16*** (0.04)	-0.10*** (0.04)
% chg. rel. to mean	-16.2	-11.4
Mean	0.98	0.90
Baseline controls	Y	Y
Observations	2484	2484

Notes: The table presents estimates of the effect of the MCH-FP on 2012–2014 outcomes at the MHSS1 household-level. Variable means refer to the comparison group. Standard errors are clustered by pre-program village. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

number of acres owned by the average household. Land ownership includes both plots used for agricultural and non-agricultural purposes. There are statistically insignificant changes of -0.04 acres (2.7 percent) in the medium-run and +0.02 acres (1.3 percent) in the long-run. Moreover, land transactions outside the family are relatively rare—fewer than 6% of households reported engaging in them in the survey year of MHSS1—and such transactions are typically small in size.¹³ These findings align with Muslim inheritance practices in Matlab, under which property is divided equally among sons and typically not sold.

The program also reduced birth rates and, in turn, population, net of improvements in childhood survival rates. Prior work documents significant fertility declines following the MCH-FP (Fauveau 1994; Joshi and Schultz 2013; Barham et al. 2023). Consistent with earlier research and Figure 3, we also show in Table 3 that the program reduced household size. Specifically, the program reduced the number of males per household aged 24 to 34 as of 2014 by 16 percent, and decreased the number of females per household in the same age range by 11 percent.¹⁴

To explore whether household size is an important margin behind the long-run sectoral reallocation documented in Table 1, we estimate a household-level instrumental variable specification that relates sectoral work allocation in 2012–2014 to the number of males aged

¹³Land transaction data in MHSS1 do not specify whether plots are agricultural or non-agricultural.

¹⁴Effect sizes are slightly smaller than in Joshi and Schultz (2013) and Barham et al. (2021) because our household-level estimation includes households less likely to have children. We include households, for example, with exclusively older individuals in the MHSS1 who had no children, and this drives down the average effect we estimate.

24–34 in the household:

$$Y_h = \alpha_0 + \alpha_1 \text{Num. males age 24 to 34}_h + \gamma X_h + \epsilon_h. \quad (14)$$

We focus on males because labor market attachment is substantially stronger for men than for women in this setting. We instrument the number of males aged 24–34 using village treatment assignment. This exercise should be interpreted as a mechanism-oriented IV scaling of the program’s reduced-form effects, rather than as identifying the structural effect of adding one more male to the household. The MCH-FP program plausibly affected later sectoral outcomes through channels beyond male cohort size, including health, schooling, and migration. We therefore use these results to assess whether the sign and magnitude of the long-run program effects are broadly consistent with a land-congestion channel.

Table 4 reports the corresponding IV-scaled estimates. Panel A reports estimates for the *share* of hours worked by sector, while Panel B presents results for the *level* of hours worked by sector. Households with program-induced reductions in the number of young adult men allocate a larger share of work time to agriculture and a smaller share to manufacturing. Turning to levels, the program has a null effect on agriculture hours, while manufacturing and services hours rise significantly with household size, in line with the marginal son being employed outside agriculture. This pattern is consistent with the model’s land-congestion mechanism: with land fixed, agricultural labor is largely pinned down,¹⁵ so reductions in the labor force raise agriculture’s employment share mainly by shrinking employment outside agriculture.

4.2 Human Capital

The second key mechanism we study is human capital. In the model, human capital raises productivity outside agriculture, increasing the nonagricultural wage and reducing the agricultural employment share, as shown in equation (5). We examine this channel using the rollout of the MCH-FP program’s child health component, which produced human-capital gains that varied across birth cohorts.

First, we confirm that returns to human capital are in fact higher outside of agriculture in this context consistent with findings by Caselli and Coleman (2001) and Porzio et al. (2022). We estimate a sector-specific Mincer equation relating log wages to years of education and potential experience. Results are presented in Appendix Table C.10. Average returns to an additional year of education in Matlab are 2.5% in agriculture, 4.7% in manufacturing, and

¹⁵The model’s equation (5) implies that the level of agricultural labor L_{at} is not a function of the population size L_t .

Table 4: MCH-FP Effects on Long-Run Sectoral Work Allocation: IV-Scaled Estimates Using Household Male Cohort Size

	(1)	(2)	(3)
	Agriculture	Manufacturing	Services
<i>Panel A: Work Time Shares by Sector</i>			
No. males aged 24–34	-0.260** (0.101)	0.201* (0.105)	0.085 (0.110)
% chg. rel. to mean	-125.8	100.2	17.5
Mean	0.21	0.20	0.48
First-stage F-stat.	14.2	14.2	14.2
Baseline controls	Y	Y	Y
Observations	2484	2484	2484
<i>Panel B: Total Work Time by Sector</i>			
No. males aged 24–34	-426 (574)	2,507*** (955)	2,001* (1,108)
% chg. rel. to mean	-31.1	141.8	49.9
Mean	1368	1769	4007
First-stage F-stat.	14.2	14.2	14.2
Baseline controls	Y	Y	Y
Observations	2484	2484	2484

Notes: The table presents 2SLS estimates for outcomes measured in the 2012–2014 MHSS2 aggregated at the MHSS1 household-level. Variable means refer to the comparison group. Standard errors are clustered by the 1996 household head’s pre-program village. The dependent variable in Panel A is the share of hours worked within the household by sector. Employment hours shares do not sum to 1 for two reasons. First, we do not report results for the construction sector. Second, we code the small set of households who do not work as spending 0 percent of their time working in each sector. The dependent variable in Panel B is total hours worked within the household by sector. See Appendix B for more details on how we classify workers into sectors. Regressions are weighted to account for household-level attrition between the MHSS1 and MHSS2 surveys. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

6.2% in services. These returns are substantially lower than the 9% average return typical of South Asia (Patrinos 2024), and are consistent with a low value of returns to education, our model’s η .

Human capital gains are concentrated among men born between 1982 and 1988, who were exposed to the program’s intensive child health phase, as documented by Barham (2012) and Barham et al. (2025) and consistent with Prediction 1 of the model. A plausible mechanism is that vaccination improves early-life health and nutrition, enhancing cognitive development and thereby increasing school attendance and learning (see Barham (2012) for additional discussion of this mechanism).

Table 5 confirms that the program’s educational gains were concentrated among men

Table 5: ITT Effects of MCH-FP on Years of Education

	(1)	(2)	(3)	(4)
	All Men	Men Born 1982–1988	Men Born 1977–1981	Men Born Pre-1977
Treatment	-0.033 (0.153)	0.737*** (0.277)	-0.428 (0.284)	0.228 (0.167)
% chg. rel. to mean	-0.6	10.5	-5.8	5.7
Mean	5.53	7.03	7.35	3.96
Baseline controls	Y	Y	Y	Y
Observations	2434	940	720	2302

Notes: The table presents estimates of the effect of the MCH-FP on 2014 outcomes at the MHSS1 household-level. Column (2) reports differences in average years of education for men born between 1982–1988. Households without men in that age range are not included in the comparison. Variable means refer to the comparison group. Standard errors are clustered by pre-program village. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

born during the intensive child health phase. For the 1982–1988 cohort, treatment increased schooling by 0.74 years, or about 11 percent relative to the comparison mean. Estimated effects for older cohorts are smaller and statistically insignificant, leaving the pooled effect across all men close to zero. Consequently, we consider the 1982–1988 birth cohort—exposed at birth to the intensive child health component of the MCH-FP—as the group that experienced significant gains in human capital relative to older cohorts.¹⁶

Following Barham et al. (2023), we exploit village treatment assignment and differential program exposure across birth cohorts using the an individual-level single-difference specification,

$$Y_i = \alpha_{y(i)} + \gamma_1(T_i \times Born_i^{77-81}) + \gamma_2(T_i \times Born_i^{82-88}) + \gamma_3(T_i \times Born_i^{Pre-77}) + \nu X_i + \epsilon_i, \quad (15)$$

where T_i is an indicator for whether i is eligible for the program as defined in Section 3.2; $\alpha_{y(i)}$ is a set of indicator variables for i 's birth year; and X_i is the vector of pre-intervention demographic and baseline characteristics shown in Appendix Table C.1.¹⁷ The outcomes Y_i we consider are the sectoral hours share and total hours worked and standard errors by the 1974 village of i (or i 's antecedents if i was not born by 1974).

¹⁶The family planning component of the program could also raise human capital through selective births. Barham et al. (2021) finds no evidence that the program changed the age at first birth, but does show a reduced age at last birth. Since family planning was available throughout the experimental period, it impacted all treated cohorts.

¹⁷We additionally control for dummy variables indicating which phase i was born in: pre-program (October 1977), first (October 1977 to February 1982), or second (March 1982 to December 1988).

γ_1 , γ_2 , and γ_3 represent the intent-to-treat single-difference coefficients of interest. γ_1 captures the effects of the family planning and maternal health interventions combined with any spillovers of having younger siblings exposed to the intensive child health interventions, and γ_2 is the combined effect of all program interventions, including the childhood health interventions which led to increases in human capital. γ_3 captures any spillover effects of the program onto older generations born prior to the intervention rollout, such as having fewer siblings.

Table 6 reports results at the individual level for men on hours worked by sector.¹⁸ Consistent with the household-level estimates, on average cohorts experience an increase in the share of hours worked in agriculture (column 1) and a reduction in manufacturing (column 2).¹⁹ However, there is heterogeneity in program effects across cohorts. For men born between 1982 and 1988—who were exposed to intensive child health interventions and experienced an increase in human capital—there was no impact on work in agriculture. Rather, they were 5.3 percentage points (10 percent) more likely to work in the service sector and 7.6 percentage points (30 percent) less likely to work in manufacturing, although the effect on service-sector employment is less precisely estimated. In contrast, men born before 1982—who did not experience improvements in schooling—exhibited patterns consistent with the land congestion mechanism. Specifically, for the 1977–1981 cohort, the share of hours worked in agriculture increased by 4.6 percentage points (57 percent), and the share in manufacturing declined by 6.5 percentage points (31 percent). Overall, while land congestion dominates when cohorts are pooled, the human-capital channel operates for the cohorts exposed to intensive child health interventions.

Total hours worked are largely unchanged for men born during the intervention. Men born before 1977, however, work approximately 146 fewer hours annually, a modest 5 percent decline relative to the comparison mean. This cohort simultaneously reallocates its work-time share from services toward agriculture, suggesting that the decline may partly be driven by changes in sectoral composition.

We conduct a battery of robustness checks on our individual-level male results in Table C.12, similar to those we did for our baseline household-level analysis. The robustness analyses does not materially change our core findings. Appendix Table C.13 reports worse-case scenario and Kling-Liebman bounds for the main outcomes in the individual-level analysis. Kling-Liebman bounds are quite consistent with our baseline estimates. While worst case bounds are quite wide, they are broadly consistent with our baseline as well.

¹⁸Given women’s low labor supply, we focus on men. See Appendix Table C.11 for results on women.

¹⁹Coefficients across each row (including the construction sector excluded from the table) must sum to 0.

Table 6: ITT Effects of MCH-FP on Long-term Work Hours by Sector: Individual-Level

	Share hours by sector			
	(1)	(2)	(3)	(4)
	Agriculture	Manufacturing	Services	Hours worked
Treatment $\times$ Born 1982–1988	0.010 (0.018)	-0.076** (0.030)	0.053 (0.035)	-24.200 (87.489)
Treatment $\times$ Born 1977–1981	0.046* (0.024)	-0.065** (0.031)	-0.012 (0.036)	-64.323 (85.682)
Treatment $\times$ Born Pre-1977	0.038* (0.022)	-0.001 (0.015)	-0.036 (0.022)	-145.842** (61.918)
% chg. (1982–88)	12.6	-30.4	10.3	-0.8
% chg. (1977–81)	56.5	-30.5	-2.0	-2.0
% chg. (Pre-1977)	13.2	-0.7	-6.9	-5.1
Comparison mean (1982–88)	0.08	0.25	0.51	3040.01
Comparison mean (1977–81)	0.08	0.21	0.59	3184.98
Comparison mean (Pre-1977)	0.29	0.10	0.52	2857.40
Observations	4744	4744	4744	4744

Notes: The table presents estimates of the effect of the MCH-FP on 2014 outcomes for men at the individual level. Means by age group refer to the comparison area. Standard errors are clustered by pre-program village. Regressions are weighted to adjust for attrition between the MHSS1 and MHSS2 surveys. All variables control for the baseline controls listed in Table C.1. The dependent variable in columns (1) through (3) is the fraction of total hours worked by sector. See Appendix B for more details on how we classify workers into sectors. Employment shares do not sum to 1 for two reasons. First, we do not report results for the construction sector. Second, a small set of respondents do not work and are coded as spending 0 percent of their time working in each of the given sectors. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

4.3 Agricultural Adaptation

Households may respond to smaller family sizes by adapting agricultural practices. Appendix Figure C.2 demonstrates the program induced a shift toward crops generating higher revenue per unit of labor, suggesting adaptation to reduced labor supply. This switching from more to less labor intensive crops represents an additional adjustment margin to reductions in population growth, complementing the sectoral reallocation documented above.

To the extent that the MCH-FP improved human capital, it may have improved agricultural productivity. We do not find evidence for this channel. We proxy per-acre productivity using revenue and profit per acre. Estimating the value of output requires crop price data, which are not available at the household level in the MHSS2. To address this, we use crop price information from the Bangladesh Statistical Yearbooks for 2012–2014. These year-

Table 7: ITT Effects of MCH-FP on Revenue and Profits per Acre

	Revenue per acre		Profit per acre	
	(1)	(2)	(3)	(4)
	Min. Price	Max. Price	Min. Price	Max. Price
Treatment	34.604 (37.612)	28.261 (47.956)	18.641 (29.487)	12.298 (40.644)
% chg. rel. to mean	10.1	5.4	11.8	3.7
Mean	341.02	519.65	157.64	336.28
Baseline controls	Y	Y	Y	Y
Observations	2003	2003	2003	2003

Notes: The table presents estimates of the effect of the MCH-FP on 2014 outcomes at the MHSS1 household-level. Standard errors are clustered by pre-program village. Revenues are constructed for each crop and are equal to the total amount of the crop harvested (in kilograms) multiplied by the prevailing national price (per kilogram). Prices are derived from the national Bangladeshi statistical yearbooks 2012–2014. Minimum (maximum) prices are the minimum (maximum) price listed in the yearbook for a given year within a crop type (e.g., Paddy Aman) among all varieties of that crop type (e.g., coarse or fine). Profits are equal to revenues net of the cost of inputs (e.g., seeds, fertilizers, pesticides, irrigation, tilling, and labor for cultivation). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

books report prices at the variety level (e.g., coarse paddy boro or fine paddy boro), rather than at the aggregate crop level (e.g., paddy boro) that we see in the MHSS data. To construct crop-level prices, we take two approaches: using either the minimum or the maximum variety-specific price within each crop category. Table 7 shows our estimates of the effect of the MCH-FP on farm productivity for the subsample of households that grew crops. Columns 1 and 2 report estimates for revenue per acre, while columns 3 and 4 present results for profits per acre. Across all specifications, we cannot statistically reject a null effect.

Similarly, the adoption of new agricultural technologies may be facilitated by having more human capital (Foster and Rosenzweig 1996). To test this, we regress a dummy for whether a household used high-yield variety seeds on a treatment dummy, with the usual vector of baseline controls. We run this crop-by-crop. Appendix Table C.14 presents the results. We find no effect of the program on the adoption of high-yield variety seeds.

5 Conclusion

This paper examines the effects of a family planning and child health program on structural transformation. These interventions can both reduce fertility and raise human capital, leaving the effect on agricultural employment ambiguous. Reduced land congestion increases returns to agricultural labor, while increased human capital raises returns to nonagricultural

work. The effects on per capita consumption expenditure is similarly ambiguous: fewer mouths to feed and higher human capital raises per capita expenditure, while a higher share of employment in agriculture lowers it. Our empirical evidence from Bangladesh shows that the Malthusian land congestion channel dominates in an open economy setting where imports can satisfy food demand.

The program therefore made households materially better off, raising per capita consumption, despite slowing down structural transformation. Moreover, fewer treated households sent fewer members to migrate out of Matlab. Assuming positive migration costs, the program may have plausibly raised average welfare of treated households.

Fertility decline is widely viewed as a key mechanism by which countries escape the “Malthusian trap” of high population growth and persistent poverty (Galor 2012). However, population decline may also dampen the pace of technological progress (Jones 2022). We contribute to the debate on the economic growth consequences of falling fertility by highlighting the underexplored mechanism of land congestion and empirically assessing its impact on structural transformation in open economies.

The broader implication of our results is that the effects of demographic transition on welfare and structural transformation need not move together. Fertility decline can raise per capita living standards while slowing the movement of labor out of agriculture. Complementary investments in human capital—through child health and education—can counteract land congestion and help generate both higher living standards and faster structural transformation.

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Online Appendix

A Theoretical Appendix

A.1 Equivalence between Linear Manufacturing Production Function and Cobb-Douglas with Internationally Traded Capital

Here we show that our linear manufacturing production function is a reduced-form representation of a standard constant-returns-to-scale production function with capital that is freely mobile across countries. Consider a manufacturing sector that produces output Q_{mt} using capital K_{mt} and effective labor $h_t L_{mt}$ according to a Cobb-Douglas technology:

$$Q_{mt} = \tilde{A}_{mt} K_{mt}^\kappa (h_t L_{mt})^{1-\kappa} \quad (\text{A.1})$$

where $\kappa \in (0, 1)$ is the capital share and $\tilde{A}_{mt}$ is total factor productivity. We assume a small open economy where capital is perfectly mobile and rented at an exogenous world interest rate r_t . Firms maximize profits taking prices p_{mt} and r_t as given. The first-order condition with respect to capital is:

$$p_{mt} \frac{\partial Q_{mt}}{\partial K_{mt}} = p_{mt} \kappa \tilde{A}_{mt} K_{mt}^{\kappa-1} (h_t L_{mt})^{1-\kappa} = r_t. \quad (\text{A.2})$$

Solving for the optimal capital stock K_{mt} :

$$K_{mt} = h_t L_{mt} \left(\frac{\kappa p_{mt} \tilde{A}_{mt}}{r_t} \right)^{\frac{1}{1-\kappa}}. \quad (\text{A.3})$$

This implies that the capital-to-effective-labor ratio is fixed. Substituting this expression back into the production function yields:

$$Q_{mt} = \underbrace{\tilde{A}_{mt} \left(\frac{\kappa p_{mt} \tilde{A}_{mt}}{r_t} \right)^{\frac{\kappa}{1-\kappa}}}_{A_{mt}} h_t L_{mt} \quad (\text{A.4})$$

Thus, output is linear in effective labor, $Q_{mt} = A_{mt} h_t L_{mt}$, where A_{mt} captures productivity and international prices. Consequently, the marginal product of labor and the wage are independent of the local labor supply.

A.2 Model in which Parents Care about Child's Future Wage

In our baseline we rule out households taking into account general equilibrium feedback: i.e., that fertility and education choices affect labor supply and therefore L_t , which affects the share of work time spent in agriculture L_{at}/L_t and thus average wages $\bar{w}_t$ (recall that agricultural wages are lower than in manufacturing by a factor of ξ). We extend our baseline model to allow for such general equilibrium forces.

Parents derive utility from their children's future wages, but take the future agricultural employment share as given. Since

$$\bar{w}_t \equiv \frac{w_{m,t}L_{m,t} + w_{a,t}L_{a,t}}{L_t} = w_{m,t}(1 - (1 - \xi)s_{a,t}),$$

this is equivalent to the households only caring about the manufacturing wage w_{mt} and hence their child's human capital.

When parents take into account general equilibrium feedback forces, optimal choices of education and fertility, equations (6) and (7), are unchanged. The key difference comes through the comparative statics, which now feature an added general equilibrium term.

Starting with per-adult labor supply, we have the following partial derivatives:

$$\begin{aligned}\frac{\partial \ell_t}{\partial p_{ft}} &= -\frac{(\alpha - \gamma\eta)\mu\pi_t\phi\bar{w}_t}{(1 + \delta + \alpha + \sigma)(\mu\pi_t\bar{w}_t\phi - p_{ft})^2} < 0 \\ \frac{\partial \ell_t}{\partial \pi_t} &= \frac{(\alpha - \gamma\eta)\mu p_{ft}\phi\bar{w}_t}{(1 + \delta + \alpha + \sigma)(\mu\pi_t\bar{w}_t\phi - p_{ft})^2} > 0 \\ \frac{\partial \ell_t}{\partial \bar{w}_t} &= \frac{(\alpha - \gamma\eta)\mu p_{ft}\pi_t\phi}{(1 + \delta + \alpha + \sigma)(\mu\pi_t\bar{w}_t\phi - p_{ft})^2} > 0.\end{aligned}$$

Combining $\bar{w}_t = \bar{w}_t(\ell_t)$ with $\ell_t = \ell_t(\bar{w}_t, p_{ft}, \pi_t)$ yields the equilibrium fixed-point condition

$$\ell_t = \ell_t(\bar{w}_t(\ell_t), p_{ft}, \pi_t).$$

For any policy shifter $x \in \{p_{ft}, \pi_t\}$,

$$\Delta_t \equiv 1 - \frac{d\bar{w}_t}{d\ell_t} \frac{\partial \ell_t}{\partial \bar{w}_t}, \quad \frac{d\ell_t}{dx} = \frac{\partial \ell_t / \partial x}{\Delta_t}, \quad \frac{d\bar{w}_t}{dx} = \frac{d\bar{w}_t}{d\ell_t} \frac{\partial \ell_t / \partial x}{\Delta_t}.$$

We assume $0 < \Delta_t < 1$ to ensure equilibrium existence and uniqueness. Intuitively, if wages respond too strongly to labor supply (e.g., a large wedge $1 - \xi$ and large $L_{a,t}/L_t$) and labor supply responds too strongly to wages (large sensitivity of fertility and contraception to income), the positive feedback loop can generate multiple equilibria or unstable amplification.

For fertility and education, we take log derivatives. The equilibrium semi-elasticities with respect to p_{ft} are

$$\frac{d \ln n_t}{dp_{ft}} = \frac{1}{(\mu\pi_t\bar{w}_t\phi - p_{ft})\Delta_t}, \quad \frac{d \ln e_{t+1}}{dp_{ft}} = -\frac{1}{(\mu\pi_t\bar{w}_t\phi - p_{ft})\Delta_t}.$$

With respect to child survival π_t , the equilibrium semi-elasticities are

$$\frac{d \ln n_t}{d\pi_t} = -\frac{1}{\pi_t} - \frac{p_{ft}}{\pi_t(\mu\pi_t\bar{w}_t\phi - p_{ft})\Delta_t}, \quad \frac{d \ln e_{t+1}}{d\pi_t} = \frac{p_{ft}}{\pi_t(\mu\pi_t\bar{w}_t\phi - p_{ft})\Delta_t}.$$

The effect on contemporaneous agricultural employment share of changes in p_{ft} or π_t is

$$\frac{d \ln(L_{at}/L_t)}{dx} = -\frac{d \ln L_t}{dx} = -\frac{d \ln \ell_t}{dx} = -\frac{1}{\ell_t} \frac{d \ell_t}{dx}, \quad x \in \{p_{ft}, \pi_t\}.$$

Under the (local) stability condition $1 - \frac{d\bar{w}_t}{d\ell_t} \frac{\partial \ell_t}{\partial \bar{w}_t} > 0$, the sign of $\frac{d \ell_t}{dx}$ equals the sign of $\frac{\partial \ell_t}{\partial x}$.

For the next period, assuming $d\ell_{t+1} \approx 0$, we get

$$d \ln \left(\frac{L_{a,t+1}}{L_{t+1}} \right) = -\frac{\eta}{1-\theta} d \ln e_{t+1} - d \ln(\pi_t n_t).$$

Using the closed forms, $e_{t+1}\pi_t n_t = \gamma\eta/(1+\delta+\alpha+\sigma)$ is constant, so $d \ln(\pi_t n_t) = -d \ln e_{t+1}$ even when $\bar{w}_t$ moves endogenously. Therefore,

$$\frac{d \ln(L_{a,t+1}/L_{t+1})}{dx} = \left(1 - \frac{\eta}{1-\theta} \right) \frac{d \ln e_{t+1}}{dx} = \frac{1-\theta-\eta}{1-\theta} \frac{d \ln e_{t+1}}{dx}, \quad x \in \{p_{ft}, \pi_t\}.$$

Finally, equation (8) for per-capita expenditure is unchanged under the general-equilibrium extension. Conditional on $\bar{w}_t$, the household first-order conditions are identical to those in the baseline model; the fixed-point term Δ_t enters only when taking total derivatives through the endogenous response of $\bar{w}_t$. Hence the same equilibrium expression for E_t applies, while its comparative statics incorporate the general-equilibrium responses derived above. Thus the general-equilibrium extension preserves Prediction 3: lower net fertility and higher manufacturing wages raise per-capita expenditure, while a higher agricultural employment share reduces it. Since demographic transition can move these margins in opposite directions, its overall effect on per-capita expenditure remains ambiguous.

Hence each comparative static and prediction retains the same sign as in our baseline model and $\eta = 1 - \theta$ continues to determine the cutoff at which land congestion dominates the human capital channel.

A.3 Baseline Model Comparative Statics

In this appendix, we explicitly derive the comparative statics of fertility n_t , education e_{t+1} , per-adult labor supply ℓ_t , and the agricultural employment share L_{at}/L_t with respect to the cost of contraception p_{ft} and child survival probability π_t .

A.3.1 Fertility and Education

Here we prove the model's Prediction 1, where households take prices (including the wage) as given. We show the results are qualitatively identical if we allow for general equilibrium adjustment to wages in Appendix A.2.

Effect of p_{ft} . Differentiating equation (6) with respect to p_{ft} :

$$\frac{\partial n_t}{\partial p_{ft}} = \frac{(\alpha - \eta\gamma)\mu\bar{w}_t}{(1 + \delta + \alpha + \sigma)(\mu\pi_t\bar{w}_t\phi - p_{ft})^2} > 0,$$

since we assumed $\alpha > \eta\gamma$. Furthermore, $\frac{\partial \pi_t n_t}{\partial p_{ft}} = \pi_t \frac{\partial n_t}{\partial p_{ft}} > 0$, so net fertility also rises with the cost of family planning.

Differentiating equation (7) with respect to p_{ft} :

$$\frac{\partial e_{t+1}}{\partial p_{ft}} = \frac{\eta\gamma}{\pi_t(\alpha - \eta\gamma)\mu\bar{w}_t} \cdot (-1) < 0.$$

Effect of π_t . Differentiating n_t with respect to π_t :

$$\frac{\partial n_t}{\partial \pi_t} = -\frac{(\alpha - \eta\gamma)\mu^2\bar{w}_t^2\phi}{(1 + \delta + \alpha + \sigma)(\mu\pi_t\bar{w}_t\phi - p_{ft})^2} < 0.$$

Turning to the effect on net fertility,

$$\frac{\partial \pi_t n_t}{\partial \pi_t} = n_t + \pi_t \frac{\partial n_t}{\partial \pi_t} < 0.$$

The sign is negative because

$$\frac{1}{\pi_t} < \frac{\partial n_t / \partial \pi_t}{n_t} = \frac{\mu\bar{w}_t}{\mu\pi_t\bar{w}_t\phi - p_{ft}} \iff \underbrace{(\phi - 1)}_{<0} \mu\pi_t\bar{w}_t < p_{ft}$$

, where ϕ , the time allocation per child must be less than 1 since total time is equal to 1.

To see why the fertility response to improved survival more than offsets the mechanical

increase in survival, note that

$$\frac{\partial \ln n_t}{\partial \ln \pi_t} = -\frac{\mu\pi_t\bar{w}_t\phi}{\mu\pi_t\bar{w}_t\phi - p_{ft}} < -1.$$

Rewriting $e_{t+1} = \frac{\eta\gamma}{\alpha - \eta\gamma} \left(\phi - \frac{p_{ft}}{\mu\bar{w}_t\pi_t} \right)$ and differentiating with respect to π_t :

$$\frac{\partial e_{t+1}}{\partial \pi_t} = \frac{\eta\gamma p_{ft}}{(\alpha - \eta\gamma)\mu\bar{w}_t\pi_t^2} > 0.$$

A.3.2 Labor Supply

Per-adult labor supply is given by:

$$\ell_t = 1 - (\phi + e_{t+1})\pi_t n_t. \quad (\text{A.5})$$

Substituting in the equilibrium values for n_t and e_{t+1} (equations (6) and (7)) simplifies the term

$$e_{t+1}\pi_t n_t = \frac{\eta\gamma}{1 + \delta + \alpha + \sigma} \quad (\text{A.6})$$

That is, total time spent on education is constant, depending only on exogenous preference parameters. Per-adult labor supply is then:

$$\ell_t = 1 - \frac{\eta\gamma}{1 + \delta + \alpha + \sigma} - \phi\pi_t n_t. \quad (\text{A.7})$$

Effect of decrease in p_{ft} . Since $\frac{\partial n_t}{\partial p_{ft}} > 0$ and π_t is constant, $\frac{\partial(\phi\pi_t n_t)}{\partial p_{ft}} > 0$. Therefore,

$$\frac{\partial \ell_t}{\partial p_{ft}} < 0. \quad (\text{A.8})$$

Effect of increase in π_t .

$$\begin{aligned} \frac{\partial \ell_t}{\partial \pi_t} &= -\phi n_t - \phi\pi_t \frac{\partial n_t}{\partial \pi_t} \\ &= \frac{p_{ft}\phi n_t}{\mu\pi_t\bar{w}_t\phi - p_{ft}} > 0, \end{aligned}$$

given our assumption that $\mu\pi_t\bar{w}_t\phi - p_{ft} > 0$, the total time endowment of 1, and the definition of ϕ as the amount of time spent raising one child.

A.3.3 Agricultural Employment Share

We consider both the contemporaneous effect (period t) and the long-run effect (period $t+1$).

Contemporaneous Effect (L_{at}/L_t). Since h_t and n_{t-1} are predetermined, the only variable in equation (5) responding to current policy is per-adult labor supply ℓ_t .

$$\frac{\partial \ln(L_{at}/L_t)}{\partial p_{ft}} = -\frac{\partial \ln L_t}{\partial p_{ft}} = -\frac{1}{\ell_t} \frac{\partial \ell_t}{\partial p_{ft}} > 0.$$

A reduction in p_{ft} increases ℓ_t , which increases the total labor force relative to land, thereby reducing the agricultural employment share via the land congestion channel. Similarly an increase in the child survival rate π_t increases per-adult labor supply, thereby reducing the agricultural employment share:

$$\frac{\partial \ln(L_{at}/L_t)}{\partial \pi_t} < 0.$$

Next-Generation Effect (L_{at+1}/L_{t+1}). Taking the log of equation (5):

$$\ln \left(\frac{L_{at+1}}{L_{t+1}} \right) = C - \frac{\eta}{1-\theta} \ln e_{t+1} - \ln L_{t+1} \quad (\text{A.9})$$

where $L_{t+1} = \pi_t n_t \ell_{t+1}$ and C collects exogenous parameters. To simplify the exposition, and consistent with our empirical findings, we assume $\frac{\partial \ell_{t+1}}{\partial p_{ft}} \approx 0$ and $\frac{\partial \ell_{t+1}}{\partial \pi_t} \approx 0$. From equation (A.6), $d \ln(\pi_t n_t) = -d \ln e_{t+1}$, changes in agricultural employment share are defined as

$$\frac{d \ln \left(\frac{L_{at+1}}{L_{t+1}} \right)}{dx} = \frac{1-\theta-\eta}{1-\theta} \frac{d \ln e_{t+1}}{dx}, \quad x \in \{p_{ft}, \pi_t\} \quad (\text{A.10})$$

Effect of decrease in p_{ft} . Since $\frac{d \ln e_{t+1}}{dp_{ft}} < 0$, $\frac{d \ln \left(\frac{L_{at+1}}{L_{t+1}} \right)}{dp_{ft}} < 0$ if and only if $1-\theta > \eta$.

Effect of increase in π_t . Since $\frac{d \ln e_{t+1}}{d\pi_t} > 0$, $\frac{d \ln \left(\frac{L_{at+1}}{L_{t+1}} \right)}{d\pi_t} > 0$ if and only if $1-\theta > \eta$.

Cross-partial derivative. Changes in p_{ft} and π_t may interact. To examine this, we look at the cross-partial derivative:

$$\frac{\partial^2 \ln(L_{a,t+1}/L_{t+1})}{\partial \pi_t \partial p_{ft}} = \frac{1-\theta-\eta}{1-\theta} \cdot \frac{\mu \bar{w}_t \phi}{(\mu \pi_t \bar{w}_t \phi - p_{ft})^2}.$$

A.3.4 Expenditure

We examine the effect of the program in period $t - 1$ on period t per-capita expenditure E_t . Given equation (8),

$$\frac{\partial E_t}{\partial \pi_t n_t} < 0,$$

$$\frac{\partial E_t}{\partial w_{mt}} > 0,$$

and

$$\frac{\partial E_t}{\partial L_{at}/L_t} < 0.$$

A.4 Partially Closed Economy

The effect of population size on structural transformation necessarily depends on whether the economy is open or closed. Our baseline model assumes a fully open economy, but the predicted effect of population size on agricultural employment share would be reversed if the economy were fully closed, as the food problem dominates. In this section, we consider the implications of nesting both closed and open economy cases by introducing trade costs.

Let τ denote iceberg trade costs, i.e., a firm must export τ units of a good in order for one unit to arrive at the destination. Then the equilibrium price of sector x 's output is

$$P_x^* = \begin{cases} P_x^{cl} & \text{if } \tau P_x^W \geq P_x^{cl} \geq P_x^W / \tau \text{ (closed)} \\ \tau P_x^W & \text{if } \tau P_x^W < P_x^{cl} \text{ (importing)} \\ P_x^W / \tau & \text{if } P_x^W / \tau > P_x^{cl} \text{ (exporting)} \end{cases} \quad (\text{A.11})$$

where P_x^W is the world price and P_x^{cl} is the prevailing local price given a closed economy. Hence whether the economy is fully closed depends on two factors: the magnitude of τ and the difference between the world price and the closed economy price.

If the agricultural sector is closed, the predicted effect of population size may reverse. A larger population induces a higher agricultural employment share in order to feed the population. If the agricultural sector imports or exports, then consistent with our baseline model a greater population induces a lower agricultural employment share.

B Classifying Industry of Employment

Neither the MHSS1 nor the MHSS2 surveys asked respondents directly about their industry of employment. Therefore, we must classify industry using indirect measures. Moreover,

because the survey questions differed between waves, we take slightly different approaches to industry classification for each survey round.

MHSS1. Employment information for MHSS1 come from three modules in the survey: (i) Book 2 Agricultural Employment (AE); (ii) Book 2 Non-Agricultural Employment (NAE); and (iii) Book 3 Employment (EMP). The household head was the respondent to the two modules from Book 2 and they provided information about household members' farm and off-farm employment. We consider all work reported in the AE module to be agricultural employment. We further classify agricultural and fishing occupations reported in NAE and EMP as agricultural employment. These occupations and their corresponding codes include: (1, 2) agriculturalist; (3) agricultural laborer; (24) fisherman; (65) husking, boiling, and drying paddy; (66) goat rearing; (67) duck or hen rearing; and (70) produce vegetables or fruits. We classify all other occupations as non-agricultural.

In each module (AE, NAE, and EMP) we observe the number of months that an individual spent working in their given occupation. To measure the amount of time an individual spends working in each sector, we sum the number of months an individual reports working in each sector across their different occupations. If that summation exceeds 12 months, we top code the amount of work in that sector at 12 months.

MHSS2. Employment information from MHSS2 come from two of the Book 3 Employment modules, Parts A and B. Part A of the Employment module (EMPA), collects information by activity type (e.g., salaried work, piece-rate work, work on the family farm) for work over the previous 12 months. From this module, we observe number of weeks worked by activity over the previous 12 months and the typical number of hours worked in that activity in a week, the product of which gives us our measure of annual hours worked in the activity.

Activities do not directly have an occupation attached to them. We assign these hours to a sector by merging the occupation code collected in Part B of the Employment module (EMPB) to the respective activity. EMPB collects information on each individual's primary and secondary occupation and each are linked to an activity from EMPA. Because only two jobs are present in EMPB, some activities in EMPA are not assigned an occupation (i.e., they worked in an activity but it was not their primary or secondary occupation). We assign work into one of four sectors: manufacturing, agriculture, service, and construction.

We classify work into manufacturing using the following rules. First, we include all factory work. We determine factory work based on whether an individual works at a government or private factory mill (empb04). We further classify factory work based on occupation codes: garment factory worker (712); jute mill worker (713); food processing factory worker

(714); and other factory machine operator (715). Finally, we rely on translated job titles and select occupation titles that include the words “factory” or “mill.” In addition to factory work, we also classify crafts-making occupations as manufacturing, including the following occupations: sheet and structural metal supervisor, moulders, and welders (621); blacksmith or tool maker (622); handicraft worker (e.g., jewelry, fabrics, pottery, printing, hand embroidery) (630); food processing (e.g., baker, butcher, dried fish maker) (650); woodworking (e.g., treaters, cabinet makers, furniture maker) (651); garment and related trade workers (e.g., tailor, seamstress, machine embroidery, upholstery, tanning) (652); other craft workers (680); and mine worker or mineral processing (711).

To classify agricultural work, we rely on the activity types that work is reported by in EMPA, as well as occupation codes from EMPB. Two activities from EMPA are explicitly related to agriculture—work as an agricultural day laborer and work on a family farm. We further include work reported in other activities in EMPA if the corresponding occupation code from EMPB is related to agricultural work: farmer, own farm (511); farmer, sharecropper (512); raising cows, goats, or sheep (513); raising ducks or hens (514); fish farm or fish hatchery (515); fishing in river or sea (516); other agriculture or forestry production (517); and agricultural laborer (820). From these, we exclude any work that was classified as manufacturing because the occupation title included the words “mill” or “factory.”

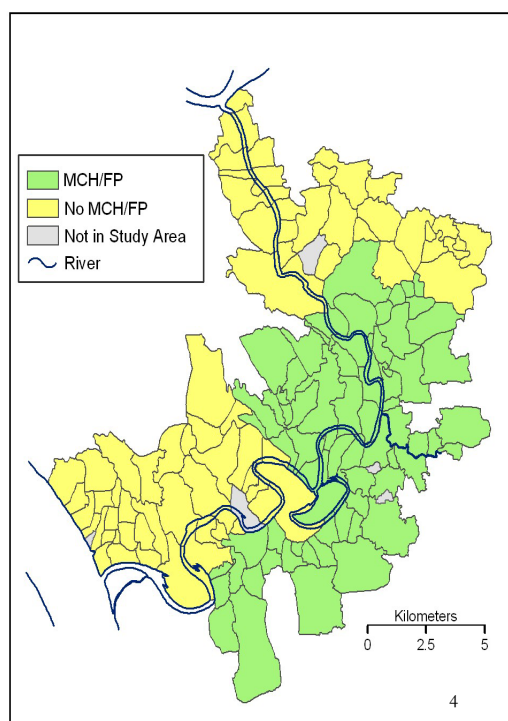
We classify a job in the service sector if the occupation corresponds to a purely service occupation, as well as other occupations not classified into the agriculture or manufacturing sector. We include occupation codes 100–442, which broadly represent work as managers (100s), professionals (200s), technicians and associate professionals (300s), and clerical support, sales workers and security (400s). Beyond these broad categories, we also include: skilled home finish or repair (612); machinery mechanics and repair (623); electrical and electronic appliance repair, maintenance and installation (640); traditional healer (660); traditional birth attendant (661); entry-level or non-degree healthcare worker (662); social worker (663); tutor (670); driver of car, van or motorcycle (730); driver of heavy equipment (731); driver of taxi, CNG, autorickshaw (732); domestic worker in home or office (811); caretaker, gardener, messenger, or doorman of home or office (812); rickshaw driver (813); boatman (814); street vendor or hawker (815); bearer or peon (816); food preparation assistant or kitchen helper (840); sweeper (860); refuse worker, sorter recycler, forager (870); and other daily laborer or elementary worker (890). We also include some records with occupation code 830 if the given occupation title was translated to be bus conductor, transport labor, tire business, transport worker, truck helper, and truck labor. Finally, any work that could not be classified with an occupation code (i.e., work reported in EMPA that did not have a corresponding job reported in EMPB) and was not in an agricultural activity was

included in service.

The final sector we classify is the construction sector. Here we include work as: carpenter, skilled house builder, supervisor, house contractor, mason (611); construction or earth-work laborer, non-food for work (821); construction or earth-work laborer, food for work (822), and any remaining unclassified work as laborer in factory, mine, or transport (830). A relatively small share of work is in construction (about 7 percent among comparison households) so we do not report results for this sector.

C Additional Tables and Figures

Figure C.1: Map of Matlab Study Area



Notes: The map plots villages in the Matlab subdistrict in Bangladesh. Villages in green are within the treatment area while those in yellow are in the comparison area.

Table C.1: Baseline Balance

	Comparison Area		Treatment Area		Difference in Means		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Mean	SD	Mean	SD	Diff.	T-stat	Diff./SD
Land size 1982 (decimals)	11.68	(15.96)	11.06	(16.19)	-0.62	-0.72	-0.04
Bari size	8.06	(5.50)	8.87	(5.99)	0.81	1.75	0.15
Family size	6.87	(2.95)	7.01	(2.94)	0.15	1.14	0.05
Wall tin or tin mix (=1)	0.314	(0.460)	0.317	(0.462)	0.003	0.13	0.01
Tin roof (=1)	0.833	(0.370)	0.828	(0.375)	-0.005	-0.26	-0.01
Number of boats	0.672	(0.623)	0.667	(0.630)	-0.006	-0.13	-0.01
Owns a lamp (=1)	0.613	(0.484)	0.652	(0.473)	0.040	1.08	0.08
Owns a watch (=1)	0.149	(0.354)	0.160	(0.364)	0.011	0.58	0.03
Owns a radio (=1)	0.080	(0.269)	0.081	(0.271)	0.001	0.10	0.00
Number of rooms (per capita)	0.206	(0.097)	0.212	(0.102)	0.007	1.49	0.07
Number of cows	1.29	(1.73)	1.45	(1.70)	0.16	1.81	0.09
Latrine (=1)	0.864	(0.341)	0.821	(0.381)	-0.043	-1.62	-0.13
Drinking water, tubewell (=1)	0.163	(0.367)	0.322	(0.464)	0.159	4.14	0.43
Drinking water, tank (=1)	0.321	(0.464)	0.394	(0.485)	0.073	1.40	0.16
HH head < 2 years education	0.610	(0.485)	0.564	(0.493)	-0.046	-1.84	-0.09
HH head works in agriculture (=1)	0.592	(0.489)	0.596	(0.487)	0.004	0.15	0.01
HH head works in fishing (=1)	0.063	(0.241)	0.055	(0.227)	-0.008	-0.49	-0.03
HH head works in business (=1)	0.096	(0.293)	0.125	(0.329)	0.029	1.40	0.10
HH head age	46.24	(13.38)	47.17	(13.72)	0.93	1.76	0.07
HH head spouse < 2 years education	0.844	(0.334)	0.806	(0.366)	-0.038	-2.02	-0.11
HH head spouse's age	36.04	(10.29)	36.65	(10.81)	0.62	1.32	0.06
1996 HH head Muslim	0.959	(0.199)	0.839	(0.367)	-0.119	-3.47	-0.60

Notes: The sample includes MHSS1 households where the household head could be traced back to the DSS area before 1977 and that had at least one household member or descendant who appeared in the MHSS2 survey. Unless otherwise noted, household characteristics come from the 1974 census. MHSS1 household baseline characteristics are traced back from the MHSS1 head. Standard deviations (SD) are clustered at the treatment village level. There are 1,176 treatment area households and 1,308 comparison area households. Standard deviations used in Column (7) come from comparison area households.

Table C.2: Attrition Balance: Individual-Level, Men and Women

	Men		Women	
	(1) Main Effect	(2) Interaction	(3) Main Effect	(4) Interaction
Constant	0.163*** (0.007)	-0.010 (0.010)	0.118*** (0.006)	0.004 (0.009)
Land size 1982 (decimals)	0.020** (0.008)	-0.024** (0.012)	0.006 (0.007)	-0.001 (0.011)
Bari size	0.002 (0.007)	0.002 (0.010)	0.010* (0.006)	-0.010 (0.009)
Family size	0.022** (0.010)	-0.025* (0.014)	0.011 (0.009)	-0.002 (0.013)
Wall tin or tin mix (=1)	0.000 (0.008)	-0.001 (0.012)	0.013* (0.007)	-0.002 (0.011)
Tin roof (=1)	0.007 (0.007)	-0.013 (0.011)	-0.006 (0.007)	0.003 (0.010)
Number of boats	-0.015* (0.008)	0.016 (0.012)	-0.003 (0.008)	0.008 (0.011)
Owens a lamp (=1)	-0.002 (0.008)	0.016 (0.011)	-0.014** (0.007)	0.020* (0.010)
Owens a watch (=1)	0.008 (0.008)	-0.016 (0.011)	0.005 (0.007)	-0.007 (0.010)
Owens a radio (=1)	-0.015** (0.008)	0.012 (0.011)	-0.014* (0.007)	0.008 (0.010)
Number of rooms (per capita)	-0.002 (0.008)	-0.005 (0.011)	0.005 (0.007)	-0.016 (0.010)
Number of cows	-0.022** (0.009)	0.034*** (0.013)	-0.014* (0.008)	-0.001 (0.012)
Latrine (=1)	-0.005 (0.007)	-0.007 (0.010)	-0.011* (0.006)	0.008 (0.009)
Drinking water, tubewell (=1)	0.001 (0.007)	0.000 (0.012)	-0.005 (0.007)	0.018* (0.011)
Drinking water, tank (=1)	-0.010 (0.007)	0.012 (0.011)	-0.006 (0.007)	0.015 (0.010)
HH head < 2 years education	0.005 (0.007)	-0.011 (0.011)	0.006 (0.007)	0.000 (0.010)
HH head works in agriculture (=1)	0.009 (0.008)	-0.015 (0.013)	0.009 (0.008)	0.001 (0.011)
HH head works in fishing (=1)	0.002 (0.007)	-0.004 (0.012)	-0.009 (0.007)	0.024** (0.011)
HH head works in business (=1)	0.004 (0.008)	-0.005 (0.012)	0.006 (0.007)	0.008 (0.011)
HH head age	-0.017 (0.012)	-0.001 (0.018)	0.002 (0.011)	-0.000 (0.016)
HH head spouse < 2 years education	-0.009 (0.007)	-0.004 (0.011)	-0.013* (0.007)	0.002 (0.010)
HH head spouse's age	0.021* (0.012)	-0.011 (0.018)	0.003 (0.011)	-0.011 (0.016)
1996 HH head Muslim	-0.019*** (0.007)	0.015 (0.011)	-0.020*** (0.006)	0.023** (0.010)
<i>F</i> -statistic that all interactions = 0		1.15		1.23
<i>p</i> -value		0.279		0.209
Observations		5,636		5,260

Notes: Each set of two columns report output from one regression where the outcome is an indicator variables that takes the value of 1 if the individual is missing MHSS2 employment information. Each regression includes the a constant term, the treatment variable, variables listed in the table, and the interaction of treatment and each of the variables listed in the table. Characteristics are standardized within sex and treatment status. The coefficient on the constant term represents the attrition rate among comparison individuals (main effect column) and the differential attrition rate for the treatment individuals (interaction column). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.3: ITT Effects of MCH-FP on Long-term Entrepreneurship and Employer Characteristics: Household-Level

	Entrepreneurship by Sector					
	(1)	(2)	(3)	(4)	(5)	(6)
	Agriculture	Manufacturing	Services	Ever worked in factory	Currently works in factory	Works at employer with > 100 employees
Treatment	0.04*** (0.01)	0.00 (0.00)	0.01 (0.01)	-0.02** (0.01)	-0.02*** (0.01)	-0.02** (0.01)
% chg. rel. to mean	19.9	5.4	3.6	-14.4	-22.6	-20.5
Mean	0.22	0.02	0.14	0.15	0.08	0.08
Baseline controls	Y	Y	Y	Y	Y	Y
Observations	2484	2484	2484	2484	2484	2484

Notes: The table presents estimates of the effect of the MCH-FP on 2014 outcomes aggregated to the MHSS1 household level. Each dependent variable is the share of household members exhibiting the described behavior. Entrepreneurship is defined as owning a business. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.4: P-values of Effect of MCH-FP on Work Time Shares by Sector: Household-Level, Inference Robustness Checks

	MHSS1 (1996)		MHSS2 (2012-2014)			
	(1) Agriculture Share	(2) Non-Agriculture Share	(3) Agriculture Share	(4) Manufacturing Share	(5) Services Share	(6) Annual Hours Per Person
Village-level cluster	0.740	0.833	0.003	0.027	0.453	0.446
Block-level cluster	0.591	0.289	0.024	0.048	0.234	0.378
Block-level wild-cluster bootstrap	0.625	0.288	0.048	0.093	0.262	0.414
Random inference, compact area	0.773	0.850	0.066	0.330	0.461	0.407
False discovery rate correction	1.000	1.000	0.019	0.072	0.829	0.829

Notes: All robustness checks are based on the estimates reported in Table 1. The table reports p-values from (i) standard errors clustered by pre-treatment village, (ii) standard errors clustered by the six pre-treatment village blocks, (iii) a wild cluster bootstrap (Roodman et al. 2019) where clusters are pre-treatment village blocks, (iv) randomization based inference where a distribution of test statistics is constructed by reassigning treatment status to villages over 10,000 permutations while maintaining a geographically contiguous and compact treatment area, and (v) adjusted p-values that control for the false discovery rate (Anderson 2008) from multiple hypothesis testing across the outcomes reported in the table using village-level clustered standard errors.

Table C.5: ITT Effects of MCH-FP on Work Time Shares by Sector: Household-Level, Robustness

	MHSS1 (1996)		MHSS2 (2012–2014)			
	(1) Agriculture Share	(2) Non-Agriculture Share	(3) Agriculture Share	(4) Manufacturing Share	(5) Services Share	(6) Annual Hours Per Person
<i>Panel A: Full Sample</i>						
Treatment	0.007 (0.021)	0.004 (0.021)	0.041*** (0.014)	-0.032** (0.014)	-0.013 (0.018)	-27.083 (35.457)
% chg. rel. to mean	1.1	1.2	19.9	-15.8	-2.8	-1.9
Mean	0.68	0.36	0.21	0.20	0.48	1445.47
Observations	2534	2534	2484	2484	2484	2484
<i>Panel B: Within 3km of Treatment Border</i>						
Treatment	-0.009 (0.027)	0.010 (0.027)	0.029* (0.017)	-0.007 (0.017)	-0.013 (0.023)	-5.758 (40.947)
% chg. rel. to mean	-1.2	3.0	13.4	-3.8	-2.8	-0.4
Mean	0.71	0.34	0.22	0.18	0.48	1425.30
Observations	1718	1718	1686	1686	1686	1686
<i>Panel C: Exclude Border Villages</i>						
Treatment	0.000 (0.026)	0.008 (0.024)	0.041** (0.017)	-0.048** (0.019)	0.002 (0.020)	-9.373 (43.720)
% chg. rel. to mean	0.0	2.3	19.5	-22.4	0.5	-0.6
Mean	0.68	0.36	0.21	0.21	0.47	1463.19
Observations	1672	1672	1640	1640	1640	1640
<i>Panel D: Only Muslim Households</i>						
Treatment	0.003 (0.022)	0.009 (0.022)	0.035** (0.015)	-0.032** (0.015)	-0.007 (0.018)	-27.852 (36.094)
% chg. rel. to mean	0.5	2.7	16.7	-16.1	-1.4	-1.9
Mean	0.68	0.35	0.21	0.20	0.48	1440.58
Observations	2286	2286	2241	2241	2241	2241
<i>Panel E: Exclude Main City</i>						
Treatment	0.018 (0.023)	-0.003 (0.023)	0.057*** (0.014)	-0.041*** (0.015)	-0.014 (0.020)	-60.979 (39.354)
% chg. rel. to mean	2.6	-0.9	27.1	-19.7	-3.1	-4.2
Mean	0.68	0.35	0.21	0.21	0.47	1446.75
Observations	2064	2064	2020	2020	2020	2020

Notes: The table presents estimates of equation (13) for outcomes at the MHSS1 household-level. Variable means refer to the comparison group. Standard errors are clustered by the 1996 household head's pre-program village. Columns (1) and (2) measure outcomes in the 1996 MHSS1, while Columns (3) through (6) measure outcomes in the 2012–2014 MHSS2. MHSS1 outcomes are the share of working months in the year in which household members could work allocated to each sector. MHSS2 outcomes are the share of hours worked by sector within the household (columns 3–5) and total average annual hours worked per person (column 6). Panel A uses the full sample of households. Panels B, C, and D restrict the sample to households from villages within 3km of the treatment border, households from non-border villages, and Muslim households, respectively. Panel E adds a control indicating whether the household's traceback village was affected by a flood protection embankment constructed in 1987. Panel F excludes households whose pre-program village is within the Matlab town boundary. See Appendix B for more details on how we classify workers into sectors. Due to changes between survey waves, sectors are constructed differently in the MHSS1 and MHSS2, and therefore are not directly comparable. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.6: ITT Effects of MCH-FP on Work Time Shares by Sector: Household-Level, By Embankment Protection Status

	MHSS1 (1996)		MHSS2 (2012–2014)			
	(1) Agriculture Share	(2) Non-Agriculture Share	(3) Agriculture Share	(4) Manufacturing Share	(5) Services Share	(6) Annual Hours Per Person
<i>Panel A: Main Specification</i>						
Treatment	0.007 (0.021)	0.004 (0.021)	0.041*** (0.014)	-0.032** (0.014)	-0.013 (0.018)	-27.083 (35.457)
% chg. rel. to mean	1.1	1.2	19.9	-15.8	-2.8	-1.9
Mean	0.68	0.36	0.21	0.20	0.48	1445.47
Observations	2534	2534	2484	2484	2484	2484
<i>Panel B: Add erosion, embankment, and distance controls</i>						
Treatment	-0.004 (0.026)	0.006 (0.027)	0.045*** (0.015)	-0.017 (0.016)	-0.013 (0.020)	-25.501 (42.414)
% chg. rel. to mean	-0.5	1.8	21.7	-8.5	-2.7	-1.8
Mean	0.68	0.36	0.21	0.20	0.48	1445.47
Observations	2534	2534	2484	2484	2484	2484
<i>Panel C: Protected vs. Unprotected</i>						
Treatment (Protected villages)	-0.005 (0.049)	0.016 (0.048)	0.066** (0.027)	0.009 (0.027)	-0.018 (0.037)	-21.179 (79.371)
Treatment (Unprotected villages)	-0.003 (0.030)	0.001 (0.031)	0.034** (0.017)	-0.031 (0.019)	-0.010 (0.025)	-27.728 (49.652)
% chg. rel. to protected mean	-0.8	4.9	33.2	4.5	-3.8	-1.5
Protected village mean	0.70	0.33	0.20	0.20	0.46	1451.02
% chg. rel. to unprotected mean	-0.39	0.29	9.00	-8.17	-2.73	-1.92
Unprotected village mean	0.66	0.37	0.37	0.37	0.37	1441.27
Observations	2534	2534	2484	2484	2484	2484

Notes: The table presents estimates of equation (13) for outcomes at the MHSS1 household-level. Variable means refer to the comparison group. Standard errors are clustered by the 1996 household head’s pre-program village. Outcomes are measured in the 2012–2014 MHSS2, and are the share of hours worked by sector within the household (columns 1–3) and total average annual hours worked per person (column 4). Panel A uses the main specification reported in Table 1. Panel B estimates effects separately by embankment protection status, where protected villages are villages where the majority of the village lies above (north and west) of the embankment from the river. Panel C further includes distance and distance-squared measures from the village centroid to the embankment, river, and icddr,b hospital. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.7: ITT Effects of MCH-FP on Work Time Shares by Sector: Household-Level, Attrition Bounds

	MHSS2 (2012–2014)			
	(1) Agriculture Share	(2) Manufacturing Share	(3) Services Share	(4) Annual Hours Per Person
<i>Panel A: Worse Case Bounds, Lower Bound</i>				
Treatment	0.020 (0.014)	-0.050*** (0.015)	-0.032* (0.017)	-117.740*** (39.621)
% chg. rel. to mean	9.1	-23.2	-6.4	-8.1
Mean	0.22	0.22	0.49	1446.37
Observations	2534	2534	2534	2534
<i>Panel B: Worse Case Bounds, Upper Bound</i>				
Treatment	0.059*** (0.014)	-0.011 (0.014)	0.008 (0.018)	69.648* (37.278)
% chg. rel. to mean	29.2	-5.5	1.6	4.8
Mean	0.20	0.20	0.47	1446.37
Observations	2534	2534	2534	2534
<i>Panel C: Kling-Liebman Bounds (+/− 0.1 SD), Lower Bound</i>				
Treatment	0.039*** (0.013)	-0.033** (0.014)	-0.013 (0.017)	-28.665 (34.702)
% chg. rel. to mean	18.7	-16.2	-2.8	-2.0
Mean	0.21	0.20	0.48	1446.37
Observations	2534	2534	2534	2534
<i>Panel D: Kling-Liebman Bounds (+/− 0.1 SD), Upper Bound</i>				
Treatment	0.041*** (0.013)	-0.030** (0.014)	-0.011 (0.017)	-22.693 (34.626)
% chg. rel. to mean	19.9	-15.2	-2.2	-1.6
Mean	0.21	0.20	0.48	1446.37
Observations	2534	2534	2534	2534

Notes: The table presents estimates of equation (13) for outcomes at the MHSS1 household-level. Variable means refer to the comparison group. Standard errors are clustered by the 1996 household head’s pre-program village. Columns (1) through (4) measure outcomes in the 2012–2014 MHSS2. MHSS2 outcomes are the share of hours worked by sector within the household (columns 1–3) and total average annual hours worked per person (column 4). Panels A and B report estimates based on worse-case-scenario bounds, assigning minimum and maximum values of outcomes to attriting households. Panels C and D report Kling-Liebman bounds, assigning attritors the mean value of the outcome ± 0.1 standard deviation, differently by treatment status. See Appendix B for more details on how we classify workers into sectors. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.8: ITT Effects of MCH-FP on Long-term Work Hour Shares by Sector and Urbanicity: Household-Level

	Urban			Rural		
	(1)	(2)	(3)	(4)	(5)	(6)
	Agriculture	Manufacturing	Services	Agriculture	Manufacturing	Services
Treatment	0.008 (0.005)	-0.037*** (0.010)	-0.015 (0.019)	0.033** (0.014)	0.005 (0.009)	0.003 (0.017)
% chg. rel. to mean	207.5	-24.0	-6.2	16.3	10.1	1.3
Mean	0.003	0.15	0.25	0.20	0.05	0.24
Baseline controls	Y	Y	Y	Y	Y	Y
Observations	2484	2484	2484	2484	2484	2484

Notes: The table presents estimates of equation (13) for outcomes measured in 2014 at the MHSS1 household-level. Variable means refer to the comparison group. Standard errors are clustered by the 1996 household head's pre-program village. The dependent variable is the share of hours worked within the household in different sectors and in different locations. See Appendix B for more details on how we classify workers into sectors. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.9: ITT Effects of MCH-FP on Land Ownership

	Acres owned	
	(1)	(2)
	MHSS1	MHSS2
	(1996)	(2012-2014)
Treatment	-0.044	0.017
	(0.108)	(0.097)
% chg. rel. to mean	-2.7	1.3
Mean	1.61	1.33
Baseline controls	Y	Y
Observations	2525	2482

Notes: The table presents estimates of equation (13) where the outcome is total acres owned by the MHSS1 household. Results for 1996 are shown in column 1, and for 2014 in column 2. Variable means refer to the comparison area. Standard errors are clustered by the 1996 household head's pre-program village. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.10: Mincer Regressions, Returns to Experience and Education by Sector

	Dependent variable: $\log(wage)$		
	(1)	(2)	(3)
	Agriculture	Manufacturing	Services
Years of education	0.023** (0.010)	0.048*** (0.008)	0.063*** (0.004)
Age	0.020 (0.024)	0.131*** (0.023)	0.059*** (0.015)
Age squared	-0.000 (0.000)	-0.002*** (0.000)	-0.001*** (0.000)
Average wage (Taka)	46	58	91
Average years of education	3.5	6.6	6.9
Average age	46.9	36.3	39.9
Observations	1129	650	2465

Notes: The table presents estimates from a Mincer wage regression by sector. The dependent variable—log hourly wage—is calculated by dividing total sectoral earnings by hours worked in the sector. The sample includes MHSS2 respondents who are MHSS1 household members or their descendants, and is restricted to men born between 1947 and 1988. Heteroskedasticity-robust standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.11: ITT Effects of MCH-FP on Long-term Work Hours by Sector: Individual-Level, Women

	Share hours by sector				
	(1)	(2)	(3)	(4)	(5)
	Agriculture	Manufacturing	Services	Non-Market	Hours worked
Treatment $\times$ Born 1982–1988	0.051*** (0.019)	0.008 (0.022)	-0.007 (0.017)	-0.052* (0.028)	46.216 (58.654)
Treatment $\times$ Born 1977–1981	-0.026 (0.032)	-0.019 (0.023)	0.030 (0.023)	0.014 (0.043)	-90.233 (83.717)
Treatment $\times$ Born Pre-1977	-0.002 (0.021)	-0.003 (0.009)	0.011 (0.009)	-0.004 (0.019)	-4.312 (29.078)
% chg. (1982–88)	40.8	6.7	-8.0	-7.8	10.9
% chg. (1977–81)	-12.5	-16.4	44.7	2.3	-17.5
% chg. (Pre-1977)	-0.6	-8.8	23.2	-0.7	-1.1
Comparison mean (1982–88)	0.12	0.12	0.08	0.67	423.77
Comparison mean (1977–81)	0.21	0.11	0.07	0.61	514.93
Comparison mean (Pre-1977)	0.36	0.03	0.05	0.56	392.90
Observations	4628	4628	4628	4628	4628

Notes: The table presents estimates of the effect of the MCH-FP on 2014 outcomes for women at the individual level. Means by age group refer to the comparison area. Standard errors are clustered by pre-program village. Regressions are weighted to adjust for attrition between the MHSS1 and MHSS2 surveys. All variables control for the baseline controls listed in Table C.1. The dependent variable in columns (1) through (3) is the fraction of total hours worked by sector. See Appendix B for more details on how we classify workers into sectors. Employment shares do not sum to 1 for two reasons. First, we do not report results for the construction sector. Second, a small set of respondents do not work and are coded as spending 0 percent of their time working in each of the given sectors. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.12: ITT Effects of MCH-FP on Long-term Work Hours by Sector: Individual-Level, Men Robustness Checks

	Share hours by sector			
	(1)	(2)	(3)	(4)
	Agriculture	Manufacturing	Services	Hours worked
<i>Panel A: Full Sample</i>				
Treatment $\times$ Born 1982–1988	0.01 (0.02)	-0.08** (0.03)	0.05 (0.04)	-24.20 (87.49)
Treatment $\times$ Born 1977–1981	0.05* (0.02)	-0.06** (0.03)	-0.01 (0.04)	-64.32 (85.68)
Treatment $\times$ Born Pre-1977	0.04* (0.02)	-0.00 (0.01)	-0.04 (0.02)	-145.84** (61.92)
Observations	4744	4744	4744	4744
<i>Panel B: Within 3km of Treatment Border</i>				
Treatment $\times$ Born 1982–1988	0.01 (0.02)	-0.05 (0.04)	0.06 (0.05)	-3.16 (100.45)
Treatment $\times$ Born 1977–1981	0.05* (0.03)	-0.04 (0.04)	0.00 (0.04)	-28.22 (110.25)
Treatment $\times$ Born Pre-1977	0.03 (0.02)	0.01 (0.02)	-0.03 (0.02)	-180.00** (84.43)
Observations	3219	3219	3219	3219
<i>Panel C: Exclude Border Villages</i>				
Treatment $\times$ Born 1982–1988	0.01 (0.02)	-0.11*** (0.04)	0.07 (0.04)	9.47 (102.55)
Treatment $\times$ Born 1977–1981	0.04 (0.03)	-0.09** (0.04)	-0.02 (0.04)	-49.94 (107.28)
Treatment $\times$ Born Pre-1977	0.02 (0.03)	-0.00 (0.02)	-0.02 (0.03)	-81.45 (70.07)
Observations	3125	3125	3125	3125

(Table continues on the next page)

	Share hours by sector			
	(1)	(2)	(3)	(4)
	Agriculture	Manufacturing	Services	Hours worked
<i>Panel D: Only Muslim Households</i>				
Treatment $\times$ Born 1982–1988	0.01 (0.02)	-0.08*** (0.03)	0.06* (0.04)	-27.60 (88.87)
Treatment $\times$ Born 1977–1981	0.05* (0.02)	-0.06* (0.03)	-0.01 (0.04)	-46.05 (89.60)
Treatment $\times$ Born Pre-1977	0.03 (0.02)	-0.00 (0.01)	-0.03 (0.02)	-156.40** (67.41)
Observations	4308	4308	4308	4308
<i>Panel E: Exclude Main Town</i>				
Treatment $\times$ Born 1982–1988	0.02 (0.02)	-0.10*** (0.03)	0.06 (0.04)	-156.40 (97.50)
Treatment $\times$ Born 1977–1981	0.05* (0.03)	-0.08** (0.03)	-0.01 (0.04)	-16.94 (102.75)
Treatment $\times$ Born Pre-1977	0.05** (0.02)	0.00 (0.01)	-0.04* (0.02)	-153.77** (64.59)
Observations	3855	3855	3855	3855
<i>Panel F: Control for Flood Embankment</i>				
Treatment $\times$ Born 1982–1988	0.01 (0.02)	-0.07** (0.03)	0.06 (0.04)	-8.29 (86.61)
Treatment $\times$ Born 1977–1981	0.05** (0.02)	-0.04 (0.03)	-0.03 (0.04)	-57.37 (88.57)
Treatment $\times$ Born Pre-1977	0.04 (0.02)	0.00 (0.01)	-0.03 (0.02)	-131.15** (65.95)
Observations	4744	4744	4744	4744

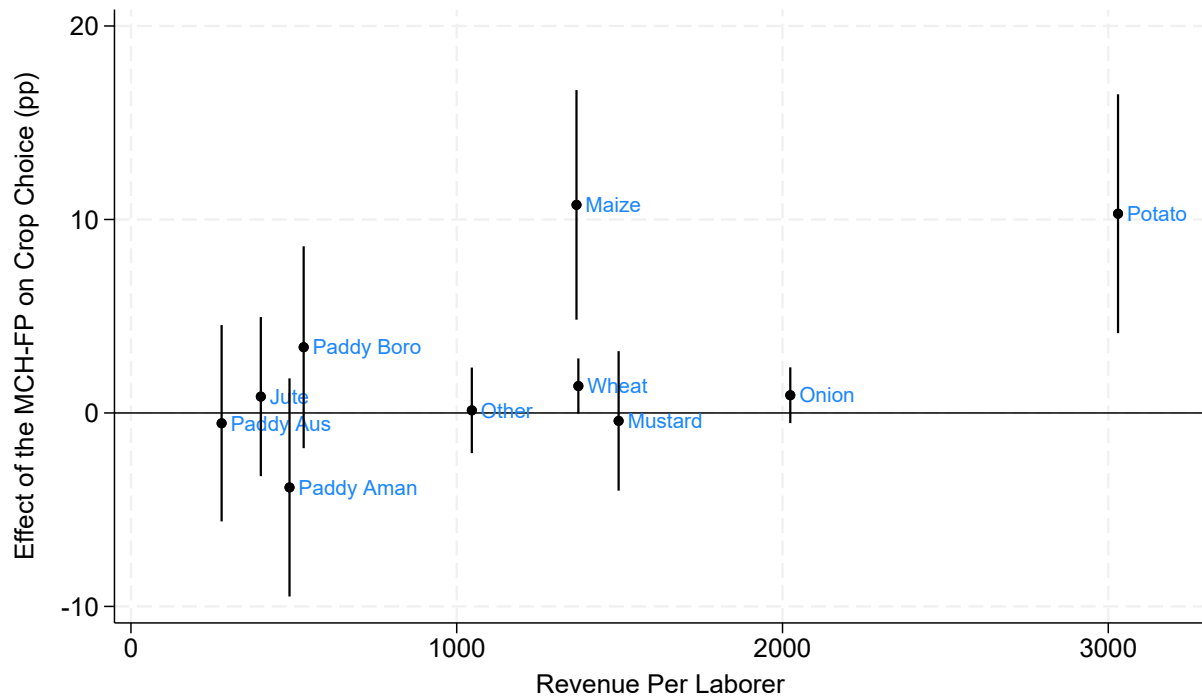
Notes: The table presents estimates of the effect of the MCH-FP on 2014 outcomes for men at the individual level. Standard errors are clustered by pre-program village. Regressions are weighted to adjust for attrition between the MHSS1 and MHSS2 surveys. All variables control for the baseline controls listed in Table C.1. The dependent variable in columns (1) through (3) is the fraction of total hours worked by sector. See Appendix B for more details on how we classify workers into sectors. Employment shares do not sum to 1 for two reasons. First, we do not report results for the construction sector. Second, a small set of respondents do not work and are coded as spending 0 percent of their time working in each of the given sectors. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.13: ITT Effects of MCH-FP on Long-term Work Hours by Sector: Individual-Level, Attrition Bounds

	Agriculture share		Manufacturing share		Services share		Hours worked	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lower bound	Upper bound	Lower bound	Upper bound	Lower bound	Upper bound	Lower bound	Upper bound
<i>Panel A: Worse Case Bounds</i>								
Treatment $\times$ Born 1982–1988	-0.09*** (0.02)	0.10*** (0.02)	-0.17*** (0.03)	0.02 (0.03)	-0.05 (0.03)	0.14*** (0.03)	-618.72*** (100.80)	576.92*** (89.51)
Treatment $\times$ Born 1977–1981	-0.10*** (0.02)	0.13*** (0.03)	-0.18*** (0.03)	0.04 (0.03)	-0.12*** (0.03)	0.10*** (0.04)	-841.29*** (110.85)	699.15*** (103.32)
Treatment $\times$ Born Pre-1977	-0.16*** (0.02)	0.22*** (0.02)	-0.20*** (0.01)	0.19*** (0.02)	-0.22*** (0.02)	0.16*** (0.02)	-1665.29*** (72.70)	1396.72*** (94.26)
Observations	5636	5636	5636	5636	5636	5636	5636	5636
<i>Panel B: Kling-Liebman Bounds (+/− 0.1 SD)</i>								
Treatment $\times$ Born 1982–1988	0.01 (0.02)	0.02 (0.02)	-0.08*** (0.03)	-0.07** (0.03)	0.04 (0.03)	0.06* (0.03)	-52.15 (79.44)	-2.41 (78.78)
Treatment $\times$ Born 1977–1981	0.04* (0.02)	0.05** (0.02)	-0.07** (0.03)	-0.05* (0.03)	-0.03 (0.03)	-0.01 (0.03)	-84.23 (76.63)	-34.00 (76.16)
Treatment $\times$ Born Pre-1977	0.02 (0.02)	0.05*** (0.02)	-0.01 (0.01)	0.01 (0.01)	-0.05*** (0.02)	-0.01 (0.02)	-180.31*** (47.32)	-81.75* (48.31)
Observations	5636	5636	5636	5636	5636	5636	5636	5636

Notes: The table presents estimates of the effect of the MCH-FP on 2014 outcomes for men at the individual level. Standard errors are clustered by pre-program village. All variables control for the baseline controls listed in Table C.1. The dependent variable in columns (1) through (6) is the fraction of total hours worked by sector. Panel A reports worse-case scenario bounds where minimum and maximum values of outcomes are assigned to attritors by cohort. Panel B reports Kling-Liebman bounds where attritors are assigned the mean value of the outcome ± 0.1 standard deviation, differently by cohort and treatment status. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure C.2: ITT Effects of MCH-FP on Crop Choice and Average Crop Productivity



Notes: The figure reports estimates of equation (13). The vertical axis reports the ITT effect of the MCH-FP on whether the household grew the given crop. The horizontal axis reports the average revenue per unit of labor when producing the crop, which comes from national Bangladeshi statistical yearbooks 2012–2014. Vertical bars represent the 95% confidence intervals.

Table C.14: Farming by Crop

	Avg. Yield (Revenue/Labor)	Grew Crop	Conditional on growing crop		
			Used HYV Seeds	Used Capital	Cost of Market Inputs
	(1)	(2)	(3)	(4)	(5)
Jute	398.76	0.008 (0.021) [0.102]	0.009 (0.064) [0.447]	-0.010 (0.019) [0.985]	-30.80 (10.75) [65.04]
Maize	1367.65	0.108*** (0.030) [0.117]	0.141 (0.043) [0.614]	0.060 (0.022) [0.934]	16.31 (12.00) [73.31]
Mustard	1496.95	-0.004 (0.018) [0.149]	0.078 (0.059) [0.311]	-0.012 (0.008) [1.000]	-7.42 (6.81) [43.43]
Onion	2023.91	0.009 (0.007) [0.017]	-0.062 (0.185) [0.523]	0.187 (0.139) [0.870]	7.16 (12.68) [12.10]
Paddy Aman	486.69	-0.038 (0.029) [0.196]	-0.072 (0.045) [0.295]	-0.020 (0.014) [0.992]	16.55 (13.17) [85.03]
Paddy Aus	278.41	-0.005 (0.026) [0.114]	-0.068 (0.051) [0.214]	0.010 (0.021) [0.980]	15.84 (9.81) [76.33]
Paddy Boro	530.16	0.034 (0.026) [0.486]	0.063 (0.042) [0.463]	0.014 (0.009) [0.976]	-3.58 (20.62) [178.66]
Potato	3029.89	0.103*** (0.031) [0.125]	0.027 (0.061) [0.423]	0.015* (0.008) [0.982]	-6.43 (64.13) [374.35]
Wheat	1373.44	0.014* (0.007) [0.008]	-0.366 (0.429) [0.500]	0.023 (0.066) [1.000]	-16.88 (17.89) [31.16]
Other	1046.34	0.001 (0.011) [0.062]	-0.022 (0.088) [0.293]	0.022 (0.032) [0.939]	-2.15 (17.50) [54.20]

Notes: The table shows various farming outcomes and choices by crop. Column 1 is computed by multiplying the average crop of the price in 2012 by the average kilograms per labor produced, with both statistics coming from Bangladesh's national agricultural yearbook. Data for columns 2 through 5 come from the MHSS2.

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